Ensemble Methods

TD – PAC-Bayesian Learning – Solutions

M2 Computer Science – MALIA

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Abstract. This tutorial accompanies Sections 2–7 of the lecture notes *PAC-Bayesian Learning: a Theoretical Framework for Ensemble Methods*. Exercises 1–3 are warm-ups (comprehension, classical short proofs, a first certificate by hand); Exercises 4–12 follow the notes; Exercises 13–17 mix applications and short proofs. Each title states the type of the exercise; those marked ($\star$) are at Master level. Notation: $S \sim \mathcal{D}^m$, voters $h : \mathcal{X} \to \{-1, +1\}$, posterior Q, prior P, Gibbs classifier G_Q, majority vote B_Q, $W_Q(x, y) = \mathbb{E}_{h\sim Q}\mathbb{1}[h(x) \neq y]$, joint error $e_Q = \mathbb{E}W_Q^2$, disagreement d_Q. Unless stated otherwise, $\delta = 0.05$, and $\overline{\mathrm{kl}}^{-1}(q, \varepsilon) = \sup\{p \in [q, 1] : \mathrm{kl}(q\|p) \leq \varepsilon\}$.

Part I – Warm-up

Exercise 1 – True or false? *(comprehension)*

For each statement, say whether it is true or false and justify your answer in one or two sentences (a short argument or a counterexample). Q and P are distributions on a set $\mathcal{H}$ of voters with values in $\{-1, +1\}$, and ties of the majority vote are counted as errors.

(a) For every posterior Q, the majority vote is at least as good as the Gibbs classifier: $R_{\mathcal{D}}(B_Q) \leq R_{\mathcal{D}}(G_Q)$.

(b) The empirical Gibbs risk $\widehat{R}_S(G_Q)$ always lies between the smallest and the largest empirical risk $\widehat{R}_S(h)$ of the voters h in the support of Q.

(c) The prior P may depend on domain knowledge or on data *other* than the sample S on which the bound is computed, but not on S itself.

(d) For the bound to be valid, the posterior Q must not depend on S either.

(e) Since a PAC-Bayesian bound holds for all posteriors simultaneously, we may compute it for a thousand posteriors and report the smallest value, without any correction.

(f) "With probability at least 0.95" means that at least 95% of the voters $h \sim P$ satisfy $R_{\mathcal{D}}(h) \leq$ the bound.

(g) For the same empirical risk $\hat{q}$ and the same complexity term ε, the kl-based bound $\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$ is never larger than the Hoeffding-type bound $\hat{q} + \sqrt{\varepsilon/2}$, and the gain is largest when $\hat{q}$ is close to 0.

(h) The disagreement d_Q can be estimated on unlabeled data, whereas the joint error e_Q needs labels.

(i) Taking $Q = P$ always gives the best bound, since it cancels the KL term.

(j) The tandem bound $4e_Q$ is always tighter than the first-order bound $2R_{\mathcal{D}}(G_Q)$.

Solution

(a) **False.** Take Q uniform on three voters and a distribution on which, on a fraction r of the points, exactly two of the three voters err, and none elsewhere. Then $W_Q = \frac{2}{3}$ on these points, the vote errs there, $R_{\mathcal{D}}(B_Q) = r$, while $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q = \frac{2r}{3} < r$. The only general relation is $R_{\mathcal{D}}(B_Q) \leq 2R_{\mathcal{D}}(G_Q)$ (Exercise 8); the vote is better only when the errors are spread over different points.

(b) **True.** $\widehat{R}_S(G_Q) = \mathbb{E}_{h\sim Q}\widehat{R}_S(h)$ is an average of the $\widehat{R}_S(h)$ with weights $Q(h)$; an average lies between the minimum and the maximum of the averaged values. (This is *not* true for the vote.)

(c) **True.** In the proof, P must be independent of S to exchange $\mathbb{E}_S$ and $\mathbb{E}_{h\sim P}$ (Fubini); any information independent of S is allowed, which is exactly what data-dependent priors learned on a separate sample exploit (Exercise 15).

(d) **False.** The bound holds, with probability $\geq 1-\delta$, *for all* Q *simultaneously*; in particular for a posterior computed from S in any way, even the minimizer of the bound (self-bounding learning, Exercise 12).

(e) **True.** It is the same event (of probability $\geq 1-\delta$ over S) that guarantees the bound for all posteriors; the minimum over a thousand of them is a data-dependent choice of Q, which is covered. Compare with (c): changing the *prior* or δ changes the event and requires a union bound.

(f) **False.** The probability is over the draw of the sample $S \sim \mathcal{D}^m$, not over voters or posteriors: for at least 95% of the samples, the bound holds for *every* posterior. Nothing is claimed about the proportion of good voters under P.

(g) **True.** By Pinsker's inequality $\mathrm{kl}(\hat{q}\|p) \geq 2(p-\hat{q})^2$, every p with $\mathrm{kl}(\hat{q}\|p) \leq \varepsilon$ satisfies $p \leq \hat{q} + \sqrt{\varepsilon/2}$, so $\overline{\mathrm{kl}}^{-1}(\hat{q},\varepsilon) \leq \hat{q} + \sqrt{\varepsilon/2}$. For $\hat{q} \to 0$ the refined Pinsker inequality gives a width of order ε instead of $\sqrt{\varepsilon}$ (Exercise 3); for $\hat{q}$ close to $\frac{1}{2}$ the two bounds nearly coincide.

(h) **True.** $d_Q = \mathbb{E}_{h,h'\sim Q}\mathbb{P}_x(h(x) \neq h'(x))$ only compares predictions, whereas "both h and h' err" requires knowing y.

(i) **False.** $Q = P$ cancels the KL, but its empirical risk is the prior average $\mathbb{E}_P\widehat{R}_S(h)$, usually large. The bound trades the two terms, and its minimizer is typically a Gibbs posterior that moves away from P (Exercises 5 and 14).

(j) **False.** Since $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$, we have $4e_Q \leq 2R_{\mathcal{D}}(G_Q)$ iff $d_Q \geq R_{\mathcal{D}}(G_Q)$. For identical voters ($d_Q = 0$), the tandem bound is $4R_{\mathcal{D}}(G_Q)$, twice the first-order bound.

Exercise 2 – From Markov to Hoeffding *(short proofs)*

In this exercise, Z is a real random variable with mean μ and variance v, and $a > 0$.

1. *(Markov).* If $Z \geq 0$, show that $\mathbb{P}(Z \geq a) \leq \frac{\mathbb{E}Z}{a}$ (hint: compare $a\mathbb{1}[Z \geq a]$ and Z). Deduce that for every $\delta \in (0,1]$, $Z \leq \frac{\mathbb{E}Z}{\delta}$ with probability at least $1-\delta$. Give a distribution for which Markov's inequality is an equality.

2. *(Chebyshev).* Show that $\mathbb{P}(|Z-\mu| \geq a) \leq \frac{v}{a^2}$.

3. *(Cantelli).* For $t \geq 0$, show that $\mathbb{P}(Z-\mu \geq a) \leq \frac{v+t^2}{(a+t)^2}$, then optimize over t to obtain $\mathbb{P}(Z-\mu \geq a) \leq \frac{v}{v+a^2}$. Compare the two inequalities numerically for $v = 0.01$ and $a = 0.2$, and explain why the one-sided bound is always better than Chebyshev's.

4. *(Union bound).* Show that $\mathbb{P}(\bigcup_{k=1}^{K} A_k) \le \sum_{k=1}^{K} \mathbb{P}(A_k)$ for any events $A_1, \dots, A_K$. Deduce that if K statements hold each with probability at least $1 - \delta/K$, then they hold *simultaneously* with probability at least $1 - \delta$.

5. *(Hoeffding's inequality).* Admit Hoeffding's lemma: if $\mathbb{E}X = 0$ and $\alpha \le X \le \beta$, then $\mathbb{E}e^{sX} \le e^{s^2(\beta-\alpha)^2/8}$ for all $s \in \mathbb{R}$. Let $Z_1, \dots, Z_m$ be i.i.d. in $[0,1]$ with mean p and $\hat{p} = \frac{1}{m}\sum_i Z_i$.

   (a) Show that $\mathbb{E}e^{\lambda(p-\hat{p})} \le e^{\lambda^2/(8m)}$ for every $\lambda > 0$.

   (b) *(Chernoff's method).* Apply Markov's inequality to $e^{\lambda(p-\hat{p})}$ and optimize over λ to show that $\mathbb{P}(p - \hat{p} \ge \varepsilon) \le e^{-2m\varepsilon^2}$.

   (c) Deduce that, with probability at least $1 - \delta$, $p \le \hat{p} + \sqrt{\ln(1/\delta)/(2m)}$, and give a two-sided version. Using question 4, give a bound valid simultaneously for K fixed classifiers.

6. ($\star$) *(Hoeffding's lemma).* Let $\alpha \le X \le \beta$ with $\mathbb{E}X = 0$ (so $\alpha < 0 < \beta$ unless $X = 0$). Using the convexity of $x \mapsto e^{sx}$, show that $\mathbb{E}e^{sX} \le e^{L(u)}$ with $u = s(\beta - \alpha)$, $\theta = \frac{-\alpha}{\beta-\alpha}$ and $L(u) = -\theta u + \ln(1 - \theta + \theta e^u)$. Show that $L(0) = L'(0) = 0$ and $L'' \le \frac{1}{4}$, and conclude.

Solution

1. If $Z \ge a$, then $a\mathbb{1}[Z \ge a] = a \le Z$; otherwise $a\mathbb{1}[Z \ge a] = 0 \le Z$. Hence $a\mathbb{1}[Z \ge a] \le Z$ pointwise, and taking expectations, $a\mathbb{P}(Z \ge a) \le \mathbb{E}Z$. With $a = \mathbb{E}Z/\delta$ (if $\mathbb{E}Z > 0$; the case $\mathbb{E}Z = 0$ is trivial since then $Z = 0$ a.s.), $\mathbb{P}(Z \ge \mathbb{E}Z/\delta) \le \delta$, i.e. $Z < \mathbb{E}Z/\delta$ with probability $\ge 1 - \delta$. *Equality:* the pointwise inequality is an equality iff $Z \in \{0, a\}$; e.g. $Z = a$ with probability r and 0 otherwise gives $\mathbb{P}(Z \ge a) = r = \mathbb{E}Z/a$. This is exactly the worst case of the first-order bound ($W_Q \in \{0, \frac{1}{2}\}$).

2. $\{|Z-\mu| \ge a\} = \{(Z-\mu)^2 \ge a^2\}$, and Markov's inequality applied to the non-negative variable $(Z-\mu)^2$ gives $\mathbb{P}(|Z-\mu| \ge a) \le \frac{\mathbb{E}(Z-\mu)^2}{a^2} = \frac{v}{a^2}$.

3. For $t \ge 0$, if $Z - \mu \ge a$ then $Z - \mu + t \ge a + t > 0$, hence $(Z - \mu + t)^2 \ge (a + t)^2$. Markov applied to $(Z - \mu + t)^2$ gives

$$\mathbb{P}(Z - \mu \ge a) \le \frac{\mathbb{E}(Z - \mu + t)^2}{(a+t)^2} = \frac{v + 2t\,\mathbb{E}(Z-\mu) + t^2}{(a+t)^2} = \frac{v + t^2}{(a+t)^2} =: g(t).$$

   Then $g'(t) = \frac{2t(a+t)-2(v+t^2)}{(a+t)^3} = \frac{2(at-v)}{(a+t)^3}$ vanishes at $t^\star = \frac{v}{a}$, is negative before and positive after. Since $v + t^{\star 2} = \frac{v(a^2+v)}{a^2}$ and $(a + t^\star)^2 = \frac{(a^2+v)^2}{a^2}$, we get $g(t^\star) = \frac{v}{v+a^2}$. *Numerically:* Chebyshev gives $\frac{0.01}{0.04} = 0.25$ (for the two-sided event), Cantelli $\frac{0.01}{0.05} = 0.20$ (for the one-sided event). Since $v + a^2 > a^2$, Cantelli is always smaller, and it is always < 1; it is also the value $g(0)$ of Chebyshev's argument that is improved by shifting the variable: the free parameter t is optimized. This is the mechanism behind the Chebyshev–Cantelli family and the C-bound (Exercise 8, question 4).

4. Pointwise, $\mathbb{1}[\bigcup_k A_k] \le \sum_k \mathbb{1}[A_k]$: if the left-hand side equals 1, at least one term of the sum equals 1. Taking expectations gives the union bound. If A_k is the event "statement k fails", then $\mathbb{P}(A_k) \le \delta/K$ and $\mathbb{P}(\text{some statement fails}) \le K \cdot \delta/K = \delta$.

5. (a) Write $p - \hat{p} = \sum_i \frac{p - Z_i}{m}$. The terms are independent, centred, with values in $[\frac{p-1}{m}, \frac{p}{m}]$, an interval of length $\frac{1}{m}$. By independence and Hoeffding's lemma,

$$\mathbb{E}e^{\lambda(p-\hat{p})} = \prod_{i=1}^{m} \mathbb{E}e^{\lambda\frac{p-Z_i}{m}} \le \prod_{i=1}^{m} e^{\lambda^2/(8m^2)} = e^{\lambda^2/(8m)}.$$

(b) For $\lambda > 0$, $x \mapsto e^{\lambda x}$ is increasing, so by Markov $\mathbb{P}(p - \hat{p} \geq \varepsilon) = \mathbb{P}(e^{\lambda(p-\hat{p})} \geq e^{\lambda\varepsilon}) \leq e^{-\lambda\varepsilon}\mathbb{E}e^{\lambda(p-\hat{p})} \leq e^{-\lambda\varepsilon+\lambda^2/(8m)}$. The exponent is minimal at $\lambda = 4m\varepsilon$, where it equals $-4m\varepsilon^2 + 2m\varepsilon^2 = -2m\varepsilon^2$. (c) Setting $e^{-2m\varepsilon^2} = \delta$ gives $\varepsilon = \sqrt{\ln(1/\delta)/(2m)}$: with probability $\geq 1-\delta$, $p < \hat{p} + \varepsilon$. The same argument applied to $1 - Z_i$ bounds $\mathbb{P}(\hat{p} - p \geq \varepsilon)$; with the union bound (each side at $\delta/2$), $|p - \hat{p}| \leq \sqrt{\ln(2/\delta)/(2m)}$ with probability $\geq 1-\delta$. For K classifiers $h_1, \ldots, h_K$ fixed *before* seeing S, applying the one-sided bound to each with δ/K gives, with probability $\geq 1-\delta$, for all k: $R_{\mathcal{D}}(h_k) \leq \widehat{R}_S(h_k) + \sqrt{\ln(K/\delta)/(2m)}$. The price of K classifiers is only logarithmic.

6. By convexity, for $x \in [\alpha, \beta]$, $e^{sx} \leq \frac{\beta-x}{\beta-\alpha}e^{s\alpha} + \frac{x-\alpha}{\beta-\alpha}e^{s\beta}$ (the chord lies above the graph). Taking expectations with $\mathbb{E}X = 0$: $\mathbb{E}e^{sX} \leq \frac{\beta e^{s\alpha} - \alpha e^{s\beta}}{\beta-\alpha} = (1-\theta)e^{s\alpha} + \theta e^{s\beta}$, with $\theta = \frac{-\alpha}{\beta-\alpha} \in [0,1]$. Since $s\alpha = -\theta u$ and $s\beta = (1-\theta)u$, this equals $e^{-\theta u}(1 - \theta + \theta e^u) = e^{L(u)}$. Now $L(0) = \ln 1 = 0$, $L'(u) = -\theta + \frac{\theta e^u}{1-\theta+\theta e^u}$ so $L'(0) = 0$, and $L''(u) = \frac{\theta e^u(1-\theta)}{(1-\theta+\theta e^u)^2} = r(1-r) \leq \frac{1}{4}$ with $r = \frac{\theta e^u}{1-\theta+\theta e^u} \in [0,1]$. By Taylor's formula with Lagrange remainder, $L(u) = L(0) + uL'(0) + \frac{u^2}{2}L''(\xi) \leq \frac{u^2}{8} = \frac{s^2(\beta-\alpha)^2}{8}$.

What we learn: every concentration inequality of the course is Markov's inequality applied to a well-chosen non-negative function (Z, $(Z-\mu)^2$, $(Z-\mu+t)^2$, $e^{\lambda(p-\hat{p})}$), followed by the optimization of a free parameter; PAC-Bayesian bounds follow the same pattern, with the change of measure in the middle.

Exercise 3 – Certifying one classifier *(application)*

A classifier f (for instance a random forest) has been learned on a sample S_1. It is then evaluated on a test sample T of $n = 200$ examples drawn i.i.d. from $\mathcal{D}$, independently of S_1, and makes 10 errors.

1. Why is $n\widehat{R}_T(f)$ a binomial variable $\mathrm{Bin}(n, R_{\mathcal{D}}(f))$ conditionally on S_1? Which assumption would break if T had been used to tune the hyperparameters of f?
2. Compute the Hoeffding test-set bound $\widehat{R}_T(f) + \sqrt{\ln(1/\delta)/(2n)}$.
3. The kl test-set bound is $\overline{\mathrm{kl}}^{-1}(\widehat{R}_T(f), \frac{\ln(1/\delta)}{n})$. Compute $\varepsilon = \ln(1/\delta)/n$, check that $\mathrm{kl}(0.05\|0.15) > \varepsilon$, and perform four steps of bisection on $[0.05, 0.15]$ (compute $\mathrm{kl}(0.05\|c)$ at each midpoint c). Compare the result with the Pinsker relaxation $\hat{q} + \sqrt{\varepsilon/2}$ and with the refined relaxation $\hat{q} + \sqrt{2\hat{q}\varepsilon} + 2\varepsilon$.
4. In fact, the engineer trained $K = 20$ classifiers (different hyperparameters), evaluated all of them on T and kept the one with the fewest test errors, which is f. Why are the certificates of questions 2 and 3 no longer valid? Correct them with a union bound, for $K = 20$ and $K = 1000$ (for the kl bound you may use `kl_inv_upper` of `pbtools`).
5. What test size n guarantees a Hoeffding width $\sqrt{\ln(K/\delta)/(2n)} \leq 0.02$, for $K \in \{1, 20, 1000\}$? Comment.
6. Show that the corrected bound of question 4 is the Occam bound (Section 2.4 of the notes) for a well-chosen prior, and give its KL term.

Solution

1. Conditionally on S_1, f is a fixed function. The test examples are i.i.d. and independent of S_1, so the indicators $\mathbb{1}[f(x_i) \neq y_i]$, $i \leq n$, are i.i.d. Bernoulli variables with parameter $\mathbb{P}_{\mathcal{D}}(f(x) \neq y) = R_{\mathcal{D}}(f)$, and their sum is binomial. If T had been used to tune f, then f would depend on T: the indicators would no longer be independent Bernoulli($R_{\mathcal{D}}(f)$) variables (the tuning selects a classifier that happens to do well

on T), and $\widehat{R}_T(f)$ would be optimistically biased.

2. $\widehat{R}_T(f) = \frac{10}{200} = 0.05$ and $\ln(1/\delta) = \ln 20 = 2.996$, so the bound is $0.05 + \sqrt{2.996/400} = 0.05 + 0.0865 = 0.1365$.

3. $\varepsilon = \frac{2.996}{200} = 0.01498$. At the right end, $\mathrm{kl}(0.05\|0.15) = 0.0507 > \varepsilon$, so $\overline{\mathrm{kl}}^{-1}(0.05, \varepsilon) \in [0.05, 0.15]$ (the function $p \mapsto \mathrm{kl}(0.05\|p)$ is increasing on $[0.05, 1]$). Bisection:

step	midpoint c	$\mathrm{kl}(0.05\|c)$	new interval
1	0.1	$0.0167 > \varepsilon$	$[0.05, 0.1]$
2	0.075	$0.0051 \leq \varepsilon$	$[0.075, 0.1]$
3	0.0875	$0.0103 \leq \varepsilon$	$[0.0875, 0.1]$
4	0.09375	$0.0134 \leq \varepsilon$	$[0.09375, 0.1]$

   After four steps we know that $\overline{\mathrm{kl}}^{-1} \in [0.094, 0.1]$, and we may already report the certified value 0.1 (the right end is always a valid upper bound). One more step ($\mathrm{kl}(0.05\|0.0969) = 0.01500 > \varepsilon$) and further iterations give $\overline{\mathrm{kl}}^{-1}(0.05, 0.01498) = 0.0968$. The Pinsker relaxation is $0.05 + \sqrt{0.00749} = 0.1365$, i.e. the Hoeffding bound of question 2, and the refined relaxation is $0.05 + \sqrt{2 \times 0.05 \times 0.01498} + 2 \times 0.01498 = 0.05 + 0.0387 + 0.0300 = 0.1187$. Hence $0.0968 < 0.1187 < 0.1365$: for a small empirical risk, the kl bound is much tighter (width 0.047 instead of 0.087).

4. Each of the K certificates is valid for a classifier fixed before looking at T; but the classifier kept is chosen *with* T, among K candidates, precisely because its test error is small. The event on which its certificate holds may be the one event that fails (a winner's curse). With the union bound (question 4 of Exercise 2), applying each certificate with δ/K makes all K of them hold simultaneously with probability $\geq 1 - \delta$, and in particular the one of the selected classifier. For $K = 20$: $\ln(K/\delta) = \ln 400 = 5.99$, Hoeffding gives $0.05 + \sqrt{5.99/400} = 0.1724$ and kl gives $\overline{\mathrm{kl}}^{-1}(0.05, 0.0300) = 0.1215$. For $K = 1000$: $\ln(20000) = 9.90$, Hoeffding $0.05 + \sqrt{9.90/400} = 0.2073$, kl $\overline{\mathrm{kl}}^{-1}(0.05, 0.0495) = 0.1484$.

5. $n \geq \frac{\ln(K/\delta)}{2 \times 0.02^2} = 1250 \ln(K/\delta)$, i.e. $n \geq 3745$ for $K = 1$, $n \geq 7490$ for $K = 20$ and $n \geq 12380$ for $K = 1000$. The precision improves like $1/\sqrt{n}$ (four times more examples to halve the width), while the number of candidates only enters through $\ln K$: comparing 1000 models "only" costs about 3.3 times more test data than certifying a single one.

6. Take the uniform prior $P = \frac{1}{K} \sum_k \delta_{f_k}$ on the K candidates and the Dirac posterior on the selected one. Then $\mathrm{KL}(\delta_f \| P) = \ln \frac{1}{P(f)} = \ln K$, and the Occam bound $R_{\mathcal{D}}(f) \leq \overline{\mathrm{kl}}^{-1}\big(\widehat{R}_T(f), \frac{\ln \frac{1}{P(f)} + \ln \frac{1}{\delta}}{n}\big)$ is exactly the corrected kl bound: the union bound is the simplest PAC-Bayesian bound, and $\ln K$ is its complexity term (Exercise 7).

What we learn: a test-set bound is only valid for a predictor chosen independently of the test sample; every data-dependent selection has a price, here $\ln K$ nats, and the kl inversion is the right tool for small risks.

Part II – Core exercises

Exercise 4 – The Kullback–Leibler divergence *(short proofs)*

Let $\mathcal{H} = \{h_1, \dots, h_n\}$ be finite and P, Q two distributions on $\mathcal{H}$ with $P_i > 0$.

1. Using Jensen's inequality, show that $\mathrm{KL}(Q\|P) \geq 0$, with equality iff $Q = P$.
2. Assume P uniform. Show that $\mathrm{KL}(Q\|P) = \ln n - H(Q)$ where $H(Q) = -\sum_i Q_i \ln Q_i$. What are the minimal and maximal values of $\mathrm{KL}(Q\|P)$? Compute it when Q is uniform over $k \leq n$ voters.
3. Show that $\mathrm{KL}(Q \otimes Q \,\|\, P \otimes P) = 2\,\mathrm{KL}(Q\|P)$.
4. Show that $\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma^2 I_d) \,\|\, \mathcal{N}(\mu_0, \sigma^2 I_d)\right) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$.
5. ($\star$) Binary kl. Fix $q \in [0,1]$. Show that $p \mapsto \mathrm{kl}(q\|p)$ is convex on $(0,1)$, minimal at $p = q$, and prove Pinsker's inequality $\mathrm{kl}(q\|p) \geq 2(q-p)^2$ by studying $g(p) = \mathrm{kl}(q\|p) - 2(q-p)^2$.

Solution

1. *Idea: $-\mathrm{KL}$ is the expectation of a logarithm, and $\ln$ is concave.* Let $I = \{i : Q_i > 0\}$ be the support of Q. By definition, $-\mathrm{KL}(Q\|P) = \sum_{i \in I} Q_i \ln \frac{P_i}{Q_i}$. Since the Q_i, $i \in I$, are weights summing to one and $\ln$ is concave, Jensen's inequality gives

$$-\mathrm{KL}(Q\|P) \leq \ln\Big(\sum_{i \in I} Q_i \frac{P_i}{Q_i}\Big) = \ln\Big(\sum_{i \in I} P_i\Big) \leq \ln 1 = 0.$$

   Equality case. Jensen's inequality for a strictly concave function is an equality iff the variable is constant: $P_i/Q_i = c$ for all $i \in I$. The second inequality is an equality iff $\sum_{i \in I} P_i = 1$, i.e. $I = \mathcal{H}$ (since all $P_i > 0$). Then $c = \sum_i P_i / \sum_i Q_i = 1$ and $Q = P$. Conversely $\mathrm{KL}(P\|P) = 0$.

2. With $P_i = \frac{1}{n}$: $\mathrm{KL}(Q\|P) = \sum_i Q_i \ln(nQ_i) = \ln n \sum_i Q_i + \sum_i Q_i \ln Q_i = \ln n - H(Q)$. The entropy satisfies $H(Q) \geq 0$ (each term $-Q_i \ln Q_i \geq 0$), with equality for a Dirac, and $H(Q) \leq \ln n$ (this is question 1 applied to $\mathrm{KL}(Q\|\text{uniform}) \geq 0$), with equality for the uniform distribution. Hence $0 \leq \mathrm{KL}(Q\|P) \leq \ln n$: the KL is 0 when the posterior is the prior and $\ln n$ when it selects a single voter (the Occam price). For Q uniform on k voters, $H(Q) = k \cdot \frac{1}{k} \ln k = \ln k$ and $\mathrm{KL} = \ln \frac{n}{k}$: selecting a tenth of the voters costs $\ln 10 \approx 2.3$ nats, whatever n.

3. Write $(Q \otimes Q)_{ij} = Q_i Q_j$. Then $\ln \frac{Q_i Q_j}{P_i P_j} = \ln \frac{Q_i}{P_i} + \ln \frac{Q_j}{P_j}$, so

$$\begin{aligned}\mathrm{KL}(Q^2\|P^2) &= \sum_{i,j} Q_i Q_j \ln \frac{Q_i}{P_i} + \sum_{i,j} Q_i Q_j \ln \frac{Q_j}{P_j} \\ &= \sum_i Q_i \ln \frac{Q_i}{P_i} + \sum_j Q_j \ln \frac{Q_j}{P_j} = 2\,\mathrm{KL}(Q\|P),\end{aligned}$$

   using $\sum_j Q_j = 1$ in the first sum and $\sum_i Q_i = 1$ in the second. This is why second-order bounds (tandem, disagreement) pay a doubled KL.

4. The two densities have the same normalizing constant, so the log-ratio is $\ln \frac{q(\theta)}{p(\theta)} = \frac{\|\theta - \mu_0\|^2 - \|\theta - \mu_1\|^2}{2\sigma^2}$. Write $\theta = \mu_1 + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_d)$ under Q. Then $\|\theta - \mu_1\|^2 = \sigma^2\|\varepsilon\|^2$ and $\|\theta - \mu_0\|^2 = \|\mu_1 - \mu_0\|^2 + 2\sigma\langle\varepsilon, \mu_1 - \mu_0\rangle + \sigma^2\|\varepsilon\|^2$. The terms in $\|\varepsilon\|^2$ cancel, and $\mathbb{E}\langle\varepsilon, \mu_1 - \mu_0\rangle = 0$, so $\mathrm{KL} = \mathbb{E}_Q \ln \frac{q}{p} = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$. Moving the centre of a Gaussian posterior costs the squared distance travelled, measured in units of the variance.

5. For $p \in (0,1)$, $\partial_p \operatorname{kl}(q\|p) = -\frac{q}{p} + \frac{1-q}{1-p} = \frac{-q(1-p)+p(1-q)}{p(1-p)} = \frac{p-q}{p(1-p)}$, which is negative for $p < q$ and positive for $p > q$: the function decreases then increases, and is minimal at $p = q$ where it vanishes. Moreover $\partial_p^2 \operatorname{kl}(q\|p) = \frac{q}{p^2} + \frac{1-q}{(1-p)^2} > 0$ (unless $q \in \{0,1\}$, where it is still ≥ 0): the function is convex. *Pinsker.* Let $g(p) = \operatorname{kl}(q\|p) - 2(q-p)^2$. Then $g(q) = 0$ and

$$g'(p) = \frac{p-q}{p(1-p)} - 4(p-q) = (p-q)\Big(\frac{1}{p(1-p)} - 4\Big).$$

Since $p(1-p) \leq \frac{1}{4}$, the bracket is non-negative, so g' has the sign of $p-q$: g decreases on $(0,q)$, increases on $(q,1)$, and its minimum is $g(q) = 0$. Hence $\operatorname{kl}(q\|p) \geq 2(q-p)^2$. The inequality is tight when $p(1-p) \approx \frac{1}{4}$, i.e. around $p = \frac{1}{2}$, which is why the kl and Hoeffding bounds coincide there.

Exercise 5 – Change of measure and Gibbs posterior *(short proofs)*

Let $\phi : \mathcal{H} \to \mathbb{R}$ and P a distribution on a finite set $\mathcal{H}$. Define $Q_\phi(h) = \frac{P(h)e^{\phi(h)}}{\mathbb{E}_P e^\phi}$.

1. Show that for any Q, $\operatorname{KL}(Q\|Q_\phi) = \operatorname{KL}(Q\|P) - \mathbb{E}_Q\phi + \ln \mathbb{E}_P e^\phi$.
2. Deduce the change of measure inequality $\mathbb{E}_Q\phi \leq \operatorname{KL}(Q\|P) + \ln \mathbb{E}_P e^\phi$ and the variational formula $\ln \mathbb{E}_P e^\phi = \max_Q\{\mathbb{E}_Q\phi - \operatorname{KL}(Q\|P)\}$.
3. Let $\lambda > 0$. Show that the minimizer of $Q \mapsto \widehat{R}_S(G_Q) + \frac{1}{\lambda m}\operatorname{KL}(Q\|P)$ is the Gibbs posterior $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$, and give the minimal value.
4. Describe Q_λ when $\lambda \to 0$ and $\lambda \to +\infty$. Interpret λ.
5. Recover Q_λ directly with a Lagrange multiplier for the constraint $\sum_h Q(h) = 1$.

Solution

1. By definition of Q_ϕ, $\ln \frac{Q(h)}{Q_\phi(h)} = \ln \frac{Q(h)}{P(h)} - \phi(h) + \ln \mathbb{E}_P e^\phi$ (the last term does not depend on h). Taking the expectation under Q: $\operatorname{KL}(Q\|Q_\phi) = \operatorname{KL}(Q\|P) - \mathbb{E}_Q\phi + \ln \mathbb{E}_P e^\phi$.
2. By Exercise 4, $\operatorname{KL}(Q\|Q_\phi) \geq 0$, hence $\mathbb{E}_Q\phi \leq \operatorname{KL}(Q\|P) + \ln \mathbb{E}_P e^\phi$ for every Q: this is the change of measure inequality. It is an equality iff $\operatorname{KL}(Q\|Q_\phi) = 0$, i.e. iff $Q = Q_\phi$. Therefore $\mathbb{E}_Q\phi - \operatorname{KL}(Q\|P) \leq \ln \mathbb{E}_P e^\phi$ for all Q, with equality at Q_ϕ: the maximum over Q is reached and equals $\ln \mathbb{E}_P e^\phi$. *Interpretation:* the left-hand side of the change of measure concerns a posterior that may depend on the data, the right-hand side only involves the prior; the price of the transfer is exactly the KL.
3. Apply question 2 to $\phi = -\lambda m \widehat{R}_S$ (a function of h once S is fixed): $-\lambda m\, \mathbb{E}_Q \widehat{R}_S - \operatorname{KL}(Q\|P) \leq \ln \mathbb{E}_P e^{-\lambda m \widehat{R}_S}$, i.e., dividing by $-\lambda m < 0$,

$$\widehat{R}_S(G_Q) + \frac{\operatorname{KL}(Q\|P)}{\lambda m} \geq -\frac{1}{\lambda m} \ln \mathbb{E}_{h\sim P} e^{-\lambda m \widehat{R}_S(h)},$$

with equality iff $Q = Q_\phi$, i.e. $Q(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$. The minimal value is the right-hand side, a "soft minimum" of the empirical risks (it tends to $\min_h \widehat{R}_S(h)$ when $\lambda \to \infty$ on a finite class).
4. When $\lambda \to 0$, $e^{-\lambda m \widehat{R}_S(h)} \to 1$ for every h and $Q_\lambda \to P$: the data are ignored. When $\lambda \to \infty$, divide numerator and denominator by $e^{-\lambda m r^\star}$ with $r^\star = \min_h \widehat{R}_S(h)$: the weight of every h with $\widehat{R}_S(h) > r^\star$ tends to 0, and Q_λ converges to P restricted to the empirical risk minimizers and renormalized. λ is an inverse temperature: it sets

how much a difference of one training error changes the relative weights (by a factor e^{λ}).

5. The objective $F(Q) = \sum_h Q(h)\widehat{R}_S(h) + \frac{1}{\lambda m}\sum_h Q(h)\ln\frac{Q(h)}{P(h)}$ is strictly convex ($x \mapsto x\ln x$ is strictly convex). With the Lagrangian $\mathcal{L}(Q,\nu) = F(Q) + \nu(\sum_h Q(h) - 1)$,

$$\frac{\partial \mathcal{L}}{\partial Q(h)} = \widehat{R}_S(h) + \frac{1}{\lambda m}\Big(\ln\frac{Q(h)}{P(h)} + 1\Big) + \nu = 0 \iff Q(h) = P(h)\, e^{-\lambda m \widehat{R}_S(h)}\, e^{-1-\lambda m\nu}.$$

The constant $e^{-1-\lambda m\nu}$ is fixed by $\sum_h Q(h) = 1$, which gives the Gibbs posterior. The positivity constraints are automatically satisfied, and by strict convexity this stationary point is the unique minimum.

Exercise 6 – From the general theorem to classical bounds *(proofs and application)*

Recall the general theorem: for convex Δ, w.p. $\geq 1-\delta$, for all Q, $\Delta(\widehat{R}_S(G_Q), R_{\mathcal{D}}(G_Q)) \leq \frac{1}{m}[\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}\mathbb{E}_{S'}\mathbb{E}_{h\sim P} e^{m\Delta(\widehat{R}_{S'}(h), R_{\mathcal{D}}(h))}]$.

1. Explain precisely where each of the following is used in the proof: convexity of Δ, the change of measure, Markov's inequality, the independence of P from S.
2. *(Hoeffding-type bound).* Take $\Delta(q,p) = \lambda(p-q)$ for a fixed $\lambda > 0$. Using Hoeffding's lemma, show that w.p. $\geq 1-\delta$, for all Q: $R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}}{\lambda m} + \frac{\lambda}{8}$.
3. Minimize the right-hand side in λ. Why is it not allowed to plug this value into the bound? Propose a correct way to optimize λ and give the resulting price.
4. *(Catoni's bound).* Let $C > 0$ and $\mathcal{F}_C(p) = -\ln(1 - p(1-e^{-C}))$. Show that for a fixed voter with risk p and the 0-1 loss, $\mathbb{E}_{S'} e^{-Cm\widehat{R}_{S'}(h)} = (1 - p(1-e^{-C}))^m$. Deduce Catoni's bound.
5. Numerical application: $m = 1000$, $\delta = 0.05$, $\widehat{R}_S(G_Q) = 0.1$, $\mathrm{KL}(Q\|P) = 5$. Compute McAllester's bound $\widehat{R}_S + \sqrt{(\mathrm{KL} + \ln\frac{2\sqrt{m}}{\delta})/(2m)}$ and the Hoeffding-type bound for $\lambda \in \{0.1, 0.2\}$.

Solution

1. The proof has three steps (Section 3.2 of the notes). *Convexity of* Δ is used in the first step (Jensen): the risks of G_Q are averages over Q of the risks of the voters, and $\Delta(\mathbb{E}_Q\widehat{R}_S(h), \mathbb{E}_Q R_{\mathcal{D}}(h)) \leq \mathbb{E}_Q\Delta(\widehat{R}_S(h), R_{\mathcal{D}}(h))$. This reduces the problem to single voters. *The change of measure* is the second step: $\mathbb{E}_Q[m\Delta] \leq \mathrm{KL}(Q\|P) + \ln\mathbb{E}_P e^{m\Delta}$. It removes Q from the random part of the inequality; since it holds for all Q at once, so will the final bound. *Markov's inequality* is the third and only probabilistic step: the non-negative variable $Z(S) = \mathbb{E}_P e^{m\Delta(\widehat{R}_S(h), R_{\mathcal{D}}(h))}$ satisfies $Z(S) \leq \mathbb{E}_{S'}Z(S')/\delta$ with probability $\geq 1-\delta$. *Independence of P and S* is used inside this step, to exchange $\mathbb{E}_{S'}$ and $\mathbb{E}_{h\sim P}$ (Fubini): for each fixed h, $\widehat{R}_{S'}(h)$ is an average of i.i.d. variables, whose exponential moment can be computed. If P depended on S', this exchange would be meaningless.
2. Here $\Delta(q,p) = \lambda(p-q)$ is linear, hence convex. For a fixed h, $\widehat{R}_{S'}(h) = \frac{1}{m}\sum_i \ell_i$ with i.i.d. $\ell_i \in [0,1]$ of mean $R_{\mathcal{D}}(h)$, so by independence

$$\mathbb{E}_{S'} e^{m\lambda(R_{\mathcal{D}}(h) - \widehat{R}_{S'}(h))} = \prod_{i=1}^{m} \mathbb{E}\, e^{\lambda(R_{\mathcal{D}}(h) - \ell_i)} \leq \big(e^{\lambda^2/8}\big)^m,$$

by Hoeffding's lemma (a centred variable with range 1 has $\mathbb{E}e^{sX} \leq e^{s^2/8}$). Hence

$\mathcal{I}_\Delta(m) \le e^{m\lambda^2/8}$, and the theorem gives $\lambda(R_\mathcal{D}(G_Q) - \widehat{R}_S(G_Q)) \le \frac{1}{m}\big[\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}\big] + \frac{\lambda^2}{8}$. Dividing by λ gives the result.

3. Let $A = \frac{\mathrm{KL}(Q\|P)+\ln(1/\delta)}{m}$. The function $\lambda \mapsto \frac{A}{\lambda} + \frac{\lambda}{8}$ is minimal at $\lambda^\star = \sqrt{8A}$, where it equals $\sqrt{A/2}$: we recover a McAllester-type bound $\widehat{R}_S(G_Q) + \sqrt{\frac{\mathrm{KL}+\ln(1/\delta)}{2m}}$. *Why it is forbidden:* λ is a parameter of Δ, i.e. of the probabilistic event on which the bound holds; this event must be fixed before drawing S. But $\lambda^\star$ depends on $\mathrm{KL}(Q\|P)$, hence on Q and on S. Using it amounts to choosing, after seeing the data, among infinitely many events, each of probability $\ge 1-\delta$: nothing guarantees that the chosen one holds. *Correct way:* fix a grid $\lambda_1 < \cdots < \lambda_k$ in advance, apply the bound to each λ_j with confidence δ/k (union bound): with probability $\ge 1-\delta$ all k bounds hold, and we can take the best one. The price is $\frac{\ln k}{\lambda m}$ (i.e. $\ln k$ added to $\ln\frac{1}{\delta}$) plus the error due to the discretization; a geometric grid with ratio 2 covering $[10^{-3}, 1]$ has $k = 10$ points and costs $\ln 10 \approx 2.3$ nats.

4. For the 0-1 loss, $m\widehat{R}_{S'}(h) \sim \mathrm{Bin}(m, p)$ with $p = R_\mathcal{D}(h)$, whose moment generating function is $\mathbb{E}e^{sB} = (1-p+pe^s)^m$. At $s = -C$: $\mathbb{E}_{S'} e^{-Cm\widehat{R}_{S'}(h)} = (1-p+pe^{-C})^m = (1-p(1-e^{-C}))^m = e^{-m\mathcal{F}_C(p)}$. Take $\Delta(q,p) = \mathcal{F}_C(p) - Cq$ (convex: $\mathcal{F}_C$ is convex, being $-\ln$ of an affine decreasing function, and $-Cq$ is linear). Then $\mathbb{E}_{S'} e^{m\Delta(\widehat{R}_{S'}(h),p)} = e^{m\mathcal{F}_C(p)} e^{-m\mathcal{F}_C(p)} = 1$, so $\mathcal{I}_\Delta(m) = 1$ and, with probability $\ge 1-\delta$, for all Q: $\mathcal{F}_C(R_\mathcal{D}(G_Q)) \le C\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)+\ln(1/\delta)}{m}$. Since $\mathcal{F}_C$ is increasing with inverse $\mathcal{F}_C^{-1}(u) = \frac{1-e^{-u}}{1-e^{-C}}$, we obtain $R_\mathcal{D}(G_Q) \le \frac{1}{1-e^{-C}}\big[1 - \exp\big(-C\widehat{R}_S(G_Q) - \frac{\mathrm{KL}+\ln(1/\delta)}{m}\big)\big]$. (For losses in $[0,1]$ that are not binary, the equality becomes an inequality by convexity of $u \mapsto e^{-Cu}$.)

5. $\ln\frac{2\sqrt{1000}}{0.05} = \ln 1264.9 \approx 7.14$, so McAllester's bound is $0.1 + \sqrt{\frac{5+7.14}{2000}} = 0.1 + 0.078 = 0.178$. For the Hoeffding-type bound, $\ln\frac{1}{0.05} \approx 3.00$ and $\mathrm{KL} + \ln\frac{1}{\delta} = 8.00$. With $\lambda = 0.1$: $0.1 + \frac{8.00}{100} + \frac{0.1}{8} = 0.1 + 0.080 + 0.0125 = 0.1925$. With $\lambda = 0.2$: $0.1 + \frac{8.00}{200} + 0.025 = 0.165$. The optimal value would be $\lambda^\star = \sqrt{8 \times 0.008} = 0.253$, giving $0.1 + \sqrt{0.004} = 0.163$. Remember that $\lambda = 0.2$ is legitimate only if it was chosen before seeing the data; if it was selected among $\{0.1, 0.2\}$ after computing both bounds, each must be computed with $\delta/2$, which gives $0.1 + \frac{5+\ln 40}{200} + 0.025 = 0.1 + 0.0434 + 0.025 \approx 0.168$ for $\lambda = 0.2$.

Exercise 7 – Occam's razor as a PAC-Bayesian bound *(comprehension)*

1. Let $\mathcal{H}$ be countable and $Q = \delta_h$ a Dirac on $h \in \mathcal{H}$. Compute $\mathrm{KL}(\delta_h\|P)$. What does Seeger's bound give for a single voter? Compare with the Occam bound of the lecture notes.
2. Why can we not use Dirac posteriors when $\mathcal{H}$ is uncountable (e.g. linear classifiers with a Gaussian prior)?
3. Let $\mathcal{H} = \{h_1, h_2, \dots\}$ be countable and $P(h_k) = \frac{6}{\pi^2 k^2}$. Check that P is a probability and give the complexity term of h_k. Interpret this prior in terms of *description length*.

Solution

1. $\mathrm{KL}(\delta_h\|P) = \sum_{h'} \delta_h(h') \ln\frac{\delta_h(h')}{P(h')} = 1 \cdot \ln\frac{1}{P(h)}$. Seeger's bound with $Q = \delta_h$ (for which

$\widehat{R}_S(G_{\delta_h}) = \widehat{R}_S(h)$) gives, simultaneously for all h,

$$R_{\mathcal{D}}(h) \le \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(h), \frac{\ln \frac{1}{P(h)} + \ln \frac{2\sqrt{m}}{\delta}}{m}\Big).$$

The Occam bound of the notes has $\ln \frac{1}{\delta}$ instead of $\ln \frac{2\sqrt{m}}{\delta}$: it is slightly tighter because it only covers Dirac posteriors (a union bound over countably many events), whereas the PAC-Bayesian bound covers all posteriors simultaneously. For $m = 1000$ the difference is $\ln(2\sqrt{1000}) \approx 4.1$ nats.

2. If P has a density (e.g. a Gaussian on $\mathbb{R}^d$), then $P(\{h\}) = 0$ for every h, and δ_h is not absolutely continuous with respect to P: $\mathrm{KL}(\delta_h \| P) = +\infty$ and the bound is void. One must use "spread" posteriors, e.g. $\mathcal{N}(\mathbf{w}, \sigma^2 I)$ around the chosen classifier; the bound then certifies the Gibbs classifier around $\mathbf{w}$, and the margin or the majority vote relates it to $\mathbf{w}$ itself (Exercise 9).

3. $\sum_{k\ge 1} \frac{1}{k^2} = \frac{\pi^2}{6}$, so $\sum_k P(h_k) = 1$ and P is a probability. The complexity of h_k is $\ln \frac{1}{P(h_k)} = 2\ln k + \ln \frac{\pi^2}{6} \approx 2\ln k + 0.50$. *Description length:* by Kraft's inequality, any prefix code assigning $\ell(h)$ nats to h defines a sub-probability $P(h) = e^{-\ell(h)}$, and conversely. The prior thus encodes a way of describing hypotheses; the bound says that a hypothesis is cheap if it has a short description *chosen before seeing the data*. Here the index k is described with about $2\ln k$ nats, so hypotheses listed early are favoured: this is Occam's razor made quantitative.

Exercise 8 – Bounds on the risk of the majority vote *(proofs and application)*

We assume voters in $\{-1, +1\}$, 0-1 loss, and count ties as errors.

1. Show that $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \ge \frac{1}{2})$ and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q$. Deduce $R_{\mathcal{D}}(B_Q) \le 2R_{\mathcal{D}}(G_Q)$. Give an example of distribution of W_Q for which this inequality is an equality.

2. Show that $e_Q = \mathbb{E}[W_Q^2]$, $d_Q = \mathbb{E}[2W_Q(1 - W_Q)]$, and $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Show also that $d_Q \le 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$.

3. Prove the tandem bound $R_{\mathcal{D}}(B_Q) \le 4e_Q$. Show that it is better than the first-order bound iff $d_Q \ge R_{\mathcal{D}}(G_Q)$.

4. For $\mu < \frac{1}{2}$, show that $R_{\mathcal{D}}(B_Q) \le \frac{\mathbb{E}(W_Q - \mu)^2}{(1/2 - \mu)^2}$. Find the optimal $\mu^\star$ and show that the optimal value is $\frac{\mathrm{Var}\, W_Q}{\mathrm{Var}\, W_Q + (1/2 - \mathbb{E}W_Q)^2}$ (Cantelli's inequality).

5. Deduce the C-bound $R_{\mathcal{D}}(B_Q) \le 1 - \frac{(1 - 2R_{\mathcal{D}}(G_Q))^2}{1 - 2d_Q}$ (when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$).

6. *(Condorcet).* Q is uniform over n voters erring independently with probability p. Compute $R_{\mathcal{D}}(G_Q)$, e_Q, d_Q as functions of n and p. For $p = 0.3$, compute the exact risk of the vote and the three bounds for $n = 3$, then the C-bound for $n = 25$ and its limit when $n \to \infty$. Comment.

Solution

1. For a fixed (x, y), $y\,\mathbb{E}_Q h(x) = \mathbb{E}_Q[yh(x)] = \mathbb{E}_Q[1 - 2\mathbb{1}[h(x) \ne y]] = 1 - 2W_Q(x, y)$, because $yh(x) = 1$ when h is right and -1 when it errs. The vote errs (ties counted as errors) iff $y\,\mathbb{E}_Q h(x) \le 0$, i.e. iff $W_Q(x, y) \ge \frac{1}{2}$. Taking the probability over $(x, y) \sim \mathcal{D}$: $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \ge \frac{1}{2})$. By Fubini, $\mathbb{E}_{\mathcal{D}} W_Q = \mathbb{E}_Q \mathbb{E}_{\mathcal{D}} \mathbb{1}[h(x) \ne y] = \mathbb{E}_Q R_{\mathcal{D}}(h) = R_{\mathcal{D}}(G_Q)$. Markov's inequality applied to $W_Q \ge 0$ gives $\mathbb{P}(W_Q \ge \frac{1}{2}) \le$

$\frac{\mathbb{E}W_Q}{1/2} = 2R_{\mathcal{D}}(G_Q)$. *Equality case:* Markov is tight when W_Q only takes the values 0 and $\frac{1}{2}$. Example: two voters with weight $\frac{1}{2}$ each, which never err at the same time, and such that on a fraction r of the points exactly one of them errs. Then $W_Q = \frac{1}{2}$ on these points, the vote errs on them (tie), $R_{\mathcal{D}}(B_Q) = r$ and $R_{\mathcal{D}}(G_Q) = \frac{r}{2}$.

2. For fixed (x, y), draw $h, h' \sim Q$ independently. The events "h errs" and "h' errs" are independent with probability W_Q each, so both err with probability W_Q^2, and exactly one errs with probability $2W_Q(1-W_Q)$; for binary voters, h and h' disagree iff exactly one of them errs. Averaging over $\mathcal{D}$ (Fubini): $e_Q = \mathbb{E}W_Q^2$ and $d_Q = \mathbb{E}[2W_Q(1-W_Q)]$. Pointwise $W_Q = W_Q^2 + \frac{1}{2}\cdot 2W_Q(1-W_Q)$, hence $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Finally Jensen (or $\operatorname{Var} W_Q \geq 0$) gives $e_Q = \mathbb{E}W_Q^2 \geq (\mathbb{E}W_Q)^2 = R_{\mathcal{D}}(G_Q)^2$, so $d_Q = 2(R_{\mathcal{D}}(G_Q) - e_Q) \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$, with equality iff W_Q is almost surely constant.

3. $\mathbb{P}(W_Q \geq \frac{1}{2}) = \mathbb{P}(W_Q^2 \geq \frac{1}{4}) \leq \frac{\mathbb{E}W_Q^2}{1/4} = 4e_Q$ (Markov applied to W_Q^2). Using question 2, $4e_Q = 4R_{\mathcal{D}}(G_Q) - 2d_Q$, so $4e_Q \leq 2R_{\mathcal{D}}(G_Q) \iff 2R_{\mathcal{D}}(G_Q) \leq 2d_Q \iff d_Q \geq R_{\mathcal{D}}(G_Q)$: the tandem bound improves on the first-order bound exactly when the voters disagree more often than an average voter errs.

4. For $\mu < \frac{1}{2}$ and $W_Q \geq \frac{1}{2}$, we have $W_Q - \mu \geq \frac{1}{2} - \mu > 0$, hence $(W_Q - \mu)^2 \geq (\frac{1}{2} - \mu)^2$: the event $\{W_Q \geq \frac{1}{2}\}$ is included in $\{(W_Q - \mu)^2 \geq (\frac{1}{2} - \mu)^2\}$ and Markov gives the bound. Write $R = \mathbb{E}W$, $e = \mathbb{E}W^2$, $v = e - R^2$ and $f(\mu) = \frac{e - 2\mu R + \mu^2}{(1/2-\mu)^2}$. Then

$$f'(\mu) = \frac{2(\mu - R)(\frac{1}{2} - \mu) + 2(e - 2\mu R + \mu^2)}{(1/2 - \mu)^3} = \frac{2(e - \frac{R}{2} - \mu(\frac{1}{2} - R))}{(1/2 - \mu)^3},$$

which vanishes at $\mu^\star = \frac{R/2 - e}{1/2 - R}$ (when $R < \frac{1}{2}$), is negative before and positive after: $\mu^\star$ is the minimum. To compute $f(\mu^\star)$, note that $\mu^\star - R = \frac{R/2 - e - R/2 + R^2}{1/2 - R} = \frac{-v}{1/2 - R}$ and $\frac{1}{2} - \mu^\star = \frac{1/4 - R/2 - R/2 + e}{1/2 - R} = \frac{v + (1/2 - R)^2}{1/2 - R}$. The numerator is $e - 2\mu^\star R + \mu^{\star 2} = (\mu^\star - R)^2 + v = \frac{v^2}{(1/2-R)^2} + v = \frac{v\,(v + (1/2-R)^2)}{(1/2-R)^2}$, so

$$f(\mu^\star) = \frac{v\,(v + (1/2 - R)^2)}{(1/2 - R)^2} \cdot \frac{(1/2 - R)^2}{(v + (1/2 - R)^2)^2} = \frac{v}{v + (1/2 - R)^2},$$

which is Cantelli's inequality for W_Q with $a = \frac{1}{2} - R$ (compare with Exercise 2, question 3).

5. With $M = 1 - 2W_Q$ (the margin), $\operatorname{Var} M = 4v$ and $\mathbb{E}M = 1 - 2R$, so $\frac{v}{v + (1/2-R)^2} = \frac{4v}{4v + (1-2R)^2} = \frac{\operatorname{Var} M}{\mathbb{E}M^2} = 1 - \frac{(\mathbb{E}M)^2}{\mathbb{E}M^2}$. Finally $\mathbb{E}M = 1 - 2R_{\mathcal{D}}(G_Q)$ and $\mathbb{E}M^2 = 1 - 4\mathbb{E}W + 4\mathbb{E}W^2 = 1 - 4R_{\mathcal{D}}(G_Q) + 4e_Q = 1 - 2d_Q$ (question 2). Hence $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1 - 2R_{\mathcal{D}}(G_Q))^2}{1 - 2d_Q}$: the C-bound is the best member of the family of question 4.

6. $nW_Q \sim \operatorname{Bin}(n, p)$ since each voter errs independently with probability p. Hence $R_{\mathcal{D}}(G_Q) = \mathbb{E}W = p$, $e_Q = \mathbb{E}W^2 = \operatorname{Var} W + p^2 = \frac{p(1-p)}{n} + p^2$, and $d_Q = 2(p - e_Q) = 2p(1-p)(1 - \frac{1}{n})$. For $p = 0.3$ and $n = 3$, the vote errs iff at least 2 voters err: $R_{\mathcal{D}}(B_Q) = 3p^2(1-p) + p^3 = 0.189 + 0.027 = 0.216$. The bounds are: first order $2p = 0.6$; $e_Q = 0.09 + 0.07 = 0.16$, so tandem $= 0.64$; $d_Q = 2 \times 0.21 \times \frac{2}{3} = 0.28$, so C-bound $= 1 - \frac{(0.4)^2}{1 - 0.56} = 1 - \frac{0.16}{0.44} \approx 0.636$. For $n = 25$: $d_Q = 0.42 \times 0.96 = 0.4032$, $1 - 2d_Q = 0.1936$ and C-bound $= 1 - \frac{0.16}{0.1936} \approx 0.174$, while the true risk is about 0.018. When $n \to \infty$, $1 - 2d_Q \to 1 - 4p(1-p) = (1-2p)^2$ and the C-bound tends to 0, like

the true risk, at the rate $\frac{4p(1-p)}{n(1-2p)^2}$. *Comment:* with 3 voters, $d_Q = 0.28 < p = 0.3$, and the second-order bounds are worse than the first-order one (Proposition 5.7 of the notes); with many independent voters, only the second-order bounds, and especially the C-bound, capture the benefit of the vote. The tandem bound tends to $4p^2 = 0.36$ and never goes to 0.

Exercise 9 – Linear classifiers with a Gaussian posterior *(proofs)*

Voters $h_{\mathbf{v}}(x) = \operatorname{sign}\langle \mathbf{v}, x\rangle$, prior $P = \mathcal{N}(\mathbf{0}, I_d)$, posterior $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$.

1. Show that $B_Q(x) = \operatorname{sign}\langle \mathbf{w}, x\rangle$.
2. Show that $\mathbb{P}_{\mathbf{v}\sim Q_{\mathbf{w}}}(h_{\mathbf{v}}(x) \neq y) = \Phi(-\frac{y\langle \mathbf{w}, x\rangle}{\|x\|})$ where Φ is the standard Gaussian CDF. Give $\widehat{R}_S(G_{Q_{\mathbf{w}}})$ and $\mathrm{KL}(Q_{\mathbf{w}}\|P)$.
3. Show that, for a fixed C, minimizing Catoni's bound over $\mathbf{w}$ amounts to minimizing $C\sum_i \Phi(-y_i\langle \mathbf{w}, x_i\rangle/\|x_i\|) + \frac{1}{2}\|\mathbf{w}\|^2$. Compare with the SVM.
4. For $\mathbf{w} = s\mathbf{u}$ with $\|\mathbf{u}\| = 1$, describe the evolution of $\widehat{R}_S(G_{Q_{s\mathbf{u}}})$ and of the KL when s grows. Relate to the notion of margin.

Solution

1. Fix x. Under $Q_{\mathbf{w}}$, $\langle \mathbf{v}, x\rangle$ is Gaussian with mean $\langle \mathbf{w}, x\rangle$ and variance $x^\top I_d x = \|x\|^2$. Hence $\mathbb{P}(\langle \mathbf{v}, x\rangle > 0) = \Phi(\frac{\langle \mathbf{w}, x\rangle}{\|x\|})$ and $\mathbb{E}_Q \operatorname{sign}\langle \mathbf{v}, x\rangle = 2\Phi(\frac{\langle \mathbf{w}, x\rangle}{\|x\|}) - 1$, which is positive iff $\langle \mathbf{w}, x\rangle > 0$ (as $\Phi(0) = \frac{1}{2}$ and Φ is increasing). Therefore $B_Q(x) = \operatorname{sign}\langle \mathbf{w}, x\rangle$: the vote of infinitely many noisy linear classifiers is the linear classifier at their centre.
2. Similarly $y\langle \mathbf{v}, x\rangle \sim \mathcal{N}(y\langle \mathbf{w}, x\rangle, \|x\|^2)$ (as $y^2 = 1$), so $\mathbb{P}_{\mathbf{v}}(h_{\mathbf{v}}(x) \neq y) = \mathbb{P}(y\langle \mathbf{v}, x\rangle \leq 0) = \Phi(-\frac{y\langle \mathbf{w}, x\rangle}{\|x\|})$. Averaging over the sample, $\widehat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m}\sum_i \Phi(-\frac{y_i\langle \mathbf{w}, x_i\rangle}{\|x_i\|})$: a smooth *probit* loss of the normalized margin $\frac{y_i\langle \mathbf{w}, x_i\rangle}{\|x_i\|}$. By Exercise 4.4 with $\sigma = 1$, $\mathrm{KL}(Q_{\mathbf{w}}\|P) = \frac{1}{2}\|\mathbf{w}\|^2$.
3. Catoni's bound reads $R_{\mathcal{D}}(G_Q) \leq \mathcal{F}_C^{-1}(C\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)+\ln(1/\delta)}{m})$, and $\mathcal{F}_C^{-1}$ is increasing: for fixed C and δ, minimizing the bound over $\mathbf{w}$ amounts to minimizing $C\widehat{R}_S(G_{Q_{\mathbf{w}}}) + \frac{\|\mathbf{w}\|^2}{2m}$, i.e. (multiplying by m) $C\sum_i \Phi(-\frac{y_i\langle \mathbf{w}, x_i\rangle}{\|x_i\|}) + \frac{1}{2}\|\mathbf{w}\|^2$. The soft-margin SVM minimizes $C\sum_i \max(0, 1 - y_i\langle \mathbf{w}, x_i\rangle) + \frac{1}{2}\|\mathbf{w}\|^2$: the regularizer is the same (it is the KL to the prior), and the hinge loss is replaced by the probit loss. Two differences: the probit loss is bounded (a badly misclassified point costs at most 1, which gives robustness to label noise) but not convex; and it uses the normalized margin.
4. For $\mathbf{w} = s\mathbf{u}$, the loss of point i is $\Phi(-s\gamma_i)$ with $\gamma_i = \frac{y_i\langle \mathbf{u}, x_i\rangle}{\|x_i\|}$. When s grows, it decreases to 0 for well-classified points ($\gamma_i > 0$), the faster the larger γ_i, and increases to 1 for misclassified points; at $s = 0$ all losses equal $\frac{1}{2}$. Meanwhile the KL grows as $\frac{s^2}{2}$. If all points have a normalized margin at least γ, choosing $s \approx \frac{c}{\gamma}$ makes the empirical Gibbs risk at most $\Phi(-c)$ at a KL cost of $\frac{c^2}{2\gamma^2}$: large margins allow small KL, which is the PAC-Bayesian margin bound (Langford and Shawe-Taylor, 2002). The trade-off in s is the one of Figure 50 of the notes (PBGD).

Exercise 10 – ($\star$) The PAC-Bayes-λ bound and its minimization *(proofs)*

1. Prove the refined Pinsker inequality $\mathrm{kl}(q\|p) \geq \frac{(p-q)^2}{2p}$ for $q < p$.
2. Starting from Seeger's bound, show that w.p. $\geq 1-\delta$, for all Q and all $\lambda \in (0,2)$ simultaneously, $R_{\mathcal{D}}(G_Q) \leq \frac{\widehat{R}_S(G_Q)}{1-\lambda/2} + \frac{\mathrm{KL}(Q\|P)+\ln\frac{2\sqrt{m}}{\delta}}{\lambda(1-\lambda/2)m}$. Why is no union bound over λ needed here, contrary to Exercise 6?
3. For a fixed Q, compute the optimal λ. For a fixed λ, compute the optimal Q on a finite set of voters. Write the alternating minimization algorithm.
4. Explain why the posterior returned by this algorithm on a random forest may give a *worse* majority vote than the uniform one.

Solution

1. Let $0 \leq q < p \leq 1$ and $g(u) = \mathrm{kl}(u\|p) - \frac{(p-u)^2}{2p}$ for $u \in [0,p]$. Then $g(p) = 0$ and $g'(u) = \ln\frac{u(1-p)}{p(1-u)} + \frac{p-u}{p}$. Using $\ln t \leq t-1$ with $t = \frac{u(1-p)}{p(1-u)}$: $\ln\frac{u(1-p)}{p(1-u)} \leq \frac{u(1-p)-p(1-u)}{p(1-u)} = \frac{u-p}{p(1-u)} \leq \frac{u-p}{p}$, the last step because $u-p<0$ and $0<1-u\leq 1$. Hence $g'(u) \leq 0$: g is non-increasing on $(0,p)$ and $g(q) \geq g(p) = 0$ (see also Appendix A.3 of the notes).
2. Let $\hat{q} = \widehat{R}_S(G_Q)$, $p = R_{\mathcal{D}}(G_Q)$ and $\varepsilon = \frac{1}{m}[\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}]$. On the event of Seeger's bound, $\mathrm{kl}(\hat{q}\|p) \leq \varepsilon$. If $p \leq \hat{q}$ the claimed bound is trivial. Otherwise the refined Pinsker inequality gives $\frac{(p-\hat{q})^2}{2p} \leq \varepsilon$, i.e. $p - \hat{q} \leq \sqrt{2p\varepsilon}$. For any $\lambda > 0$, the arithmetic–geometric inequality $\sqrt{ab} \leq \frac{a+b}{2}$ with $a = \lambda p$, $b = \frac{2\varepsilon}{\lambda}$ gives $\sqrt{2p\varepsilon} \leq \frac{\lambda p}{2} + \frac{\varepsilon}{\lambda}$. Hence $p(1-\frac{\lambda}{2}) \leq \hat{q} + \frac{\varepsilon}{\lambda}$, and for $\lambda \in (0,2)$ we can divide by $1-\frac{\lambda}{2} > 0$. *Why no union bound:* all these inequalities are deterministic consequences of the *single* event of Seeger's bound; λ appears only in the algebra, not in the probabilistic statement. In Exercise 6, on the contrary, λ defined the function Δ, hence the event itself.
3. *Optimal λ for fixed Q.* Let $B(\lambda) = \frac{\hat{q}}{1-\lambda/2} + \frac{\varepsilon}{\lambda(1-\lambda/2)}$. Setting $B'(\lambda) = 0$ leads to $\frac{\hat{q}}{2}\lambda^2 = \varepsilon(1-\lambda)$, i.e. $\hat{q}\lambda^2 + 2\varepsilon\lambda - 2\varepsilon = 0$, whose positive root is $\lambda^\star = \frac{-\varepsilon+\sqrt{\varepsilon^2+2\hat{q}\varepsilon}}{\hat{q}} = \frac{2}{\sqrt{2\hat{q}/\varepsilon+1}+1}$. *Optimal Q for fixed λ.* The bound is an increasing affine function of $\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda m}$, minimized by the Gibbs posterior $Q(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$ (Exercise 5). *Algorithm:* $\lambda \leftarrow 1$; repeat $Q \leftarrow$ Gibbs posterior at λ, then $\lambda \leftarrow \lambda^\star(Q)$, until λ stabilizes; return Q and the certificate computed with the kl form. Each step decreases the bound, which is therefore non-increasing along the iterations (and converges to the global minimum under the conditions of Thiemann et al., 2017).
4. The bound only involves the *average* OOB loss $\rho^\top \widehat{L}$ and the KL. It therefore puts the weight on the few trees with the smallest OOB loss, which tend to be similar (they are grown on similar bootstrap samples and make similar errors). The effective number of trees collapses (to about 2 on `digits`), the disagreement vanishes, and the vote loses the benefit of averaging: its risk approaches that of a single good tree (0.150 instead of 0.044 on `digits`, Table 3 of the notes). The bound is excellent for the Gibbs classifier, but it is the wrong objective for the vote.

Exercise 11 – (⋆) Out-of-bag samples *(proofs)*

Consider bagging with bootstrap samples of size k drawn with replacement among the m training examples.

1. Compute the probability that a given example is out-of-bag for a given tree; give its limit when $m \to \infty$ for $k = m$ and for $k = m/2$. Same question for being out-of-bag for two given trees.
2. Explain why the OOB loss $\widehat{L}_t$ of tree t is an unbiased estimate of $R_{\mathcal{D}}(h_t)$ given S_t, and why a uniform prior over the *indices* of the trees is a legitimate PAC-Bayesian prior although the trees depend on S.
3. Show that if $\mathbb{E}e^{N\,\mathrm{kl}(\hat{p}_N\|p)} \le 2\sqrt{N}$ for all N, then for $n_0 \le N$, $\mathbb{E}e^{n_0\,\mathrm{kl}(\hat{p}_N\|p)} \le 2\sqrt{n_0}$. Why is this useful?

Solution

1. A bootstrap sample of size k draws k indices uniformly and independently in $\{1,\dots,m\}$. A given example is never drawn with probability $(1-\frac{1}{m})^k$. Since $(1-\frac{1}{m})^m \to e^{-1}$, this tends to $e^{-1} \approx 0.368$ for $k = m$ and to $e^{-1/2} \approx 0.607$ for $k = m/2$. For two trees with independent bootstrap samples, the example is out-of-bag for both with probability $(1-\frac{1}{m})^{2k} \to e^{-2} \approx 0.135$ ($k = m$) or $e^{-1} \approx 0.368$ ($k = m/2$). Halving the bootstrap size thus almost triples the size of the pairwise OOB sets used by the tandem bound, at the price of slightly weaker trees.
2. Condition on the bootstrap sample S_t (and on the internal randomness of the tree). Then h_t is a fixed function, and the OOB examples, which were not used to build it, are still i.i.d. from $\mathcal{D}$ and independent of h_t (they are the examples of S whose index was not drawn; given the drawn indices, their values are independent of S_t). Hence $\mathbb{E}[\widehat{L}_t \,|\, S_t] = R_{\mathcal{D}}(h_t)$, and Maurer's lemma applies to $\widehat{L}_t$ conditionally on S_t. The prior is placed on the *indices* $t \in \{1,\dots,n\}$, not on the trees themselves: the uniform distribution on indices is fixed before seeing the data. The dependence of h_t on S only enters the third step of the proof, where, for each index t, the conditional concentration of $\widehat{L}_t$ is used.
3. Write $e^{n_0\,\mathrm{kl}(\hat{p}_N\|p)} = \left(e^{N\,\mathrm{kl}(\hat{p}_N\|p)}\right)^{n_0/N}$. Since $n_0 \le N$, $x \mapsto x^{n_0/N}$ is concave, and Jensen gives $\mathbb{E}e^{n_0\,\mathrm{kl}} \le \left(\mathbb{E}e^{N\,\mathrm{kl}}\right)^{n_0/N} \le (2\sqrt{N})^{n_0/N} = \exp\left(n_0 \frac{\ln(2\sqrt{N})}{N}\right)$. The function $N \mapsto \frac{\ln(2\sqrt{N})}{N}$ is decreasing for $N \ge 1$ (its derivative is $\frac{1/2 - \ln(2\sqrt{N})}{N^2} < 0$), so for $N \ge n_0$ this is at most $\exp\left(n_0 \frac{\ln(2\sqrt{n_0})}{n_0}\right) = 2\sqrt{n_0}$. *Usefulness:* the OOB sets have different sizes; this lemma allows us to use a single sample size $n_0 = \min_t |T_t|$ in the complexity term while still computing each loss $\widehat{L}_t$ on *all* the OOB examples of tree t.

Exercise 12 – (⋆) Self-bounding learning *(comprehension and proofs)*

Let $\mathcal{B}(Q, S, \delta)$ be a bound such that $\mathbb{P}_S(\forall Q,\ R_{\mathcal{D}}(Q) \le \mathcal{B}(Q, S, \delta)) \ge 1 - \delta$.

1. Let Q_S be the posterior returned by any algorithm run on S. Show that $\mathbb{P}_S(R_{\mathcal{D}}(Q_S) \le \mathcal{B}(Q_S, S, \delta)) \ge 1 - \delta$. Why does this fail for a bound that holds only for a *fixed* predictor (e.g. Hoeffding's inequality)?
2. You build $K = 5$ random forests with tree depths $1, 2, 4, 8$ and unlimited, and compute for each of them the OOB tandem certificate. How must you modify the certificates to be allowed to keep the forest with the smallest one? What is the cost in the complexity term?

3. Among the following choices, which are free, which require a union bound, which are forbidden? (a) the temperature λ of a Gibbs posterior; (b) the parameter C of Catoni's bound; (c) the variance of a Gaussian prior chosen on a grid of 10 values; (d) the centre of a Gaussian prior learned on the sample used in the bound; (e) the number of gradient iterations when minimizing the bound.
4. Let $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$. Show that $\frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. Deduce the gradient of the first-order certificate $2\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{L}, \frac{\mathrm{KL}(\rho\|\pi)+c}{n})$ with respect to ρ.
5. Why is the test-set bound usually tighter than a self-bounding certificate for a random forest? Give one advantage of the self-bounding approach.

Solution

1. Let $E = \{S : \forall Q,\ R_{\mathcal{D}}(Q) \le \mathcal{B}(Q, S, \delta)\}$, with $\mathbb{P}(E) \ge 1-\delta$. For $S \in E$ the inequality holds for every posterior, in particular for Q_S, whatever the way it was computed. Hence $\{S : R_{\mathcal{D}}(Q_S) \le \mathcal{B}(Q_S, S, \delta)\} \supseteq E$ and has probability $\ge 1-\delta$. For a fixed-predictor bound, there is one event E_h per predictor, each of probability $\ge 1-\delta$, but their intersection can have a much smaller probability. The learning algorithm, which looks at S, may precisely choose a predictor for which E_h fails: among 1000 random classifiers of risk 0.5 evaluated on 100 examples, the one with the smallest empirical risk violates the naive Hoeffding bound in 99.8% of the runs (Figure 18 of the notes).
2. Keeping the smallest certificate is a data-dependent choice among $K = 5$ bounds. By the union bound (Proposition 6.3 of the notes), if each certificate is computed with confidence $\delta/5$, all five hold simultaneously with probability $\ge 1-\delta$, and so does the smallest one. For the tandem bound, the complexity term $\frac{2\,\mathrm{KL} + \ln(2\sqrt{n_0}/\delta)}{n_0}$ becomes $\frac{2\,\mathrm{KL} + \ln(2\sqrt{n_0}/\delta) + \ln 5}{n_0}$: the price is $\ln 5 \approx 1.6$ nats, i.e. 0.004 in the complexity term for $n_0 = 400$.
3. (a) *Free*: a temperature only indexes a family of posteriors compared with the same prior in the same bound. (b) *Union bound*: C defines the function Δ, hence the event; use a grid fixed in advance with δ/k. (c) *Union bound*: each variance defines a different prior, hence a different bound; cost $\ln 10$. (d) *Forbidden*: the prior would depend on the sample used in the bound, and step 3 of the proof fails; it becomes legitimate if the centre is learned on a separate part S_1 and the bound computed on S_2 (Proposition 6.5). (e) *Free*: any stopping rule produces a posterior, and the bound holds for all posteriors.
4. Let $F(q, p) = \mathrm{kl}(q\|p) - \varepsilon$, so that $F(q, \overline{\mathrm{kl}}^{-1}(q, \varepsilon)) = 0$. We have $F_q = \partial_q \mathrm{kl}(q\|p) = \ln \frac{q(1-p)}{p(1-q)}$ and $F_p = \frac{p-q}{p(1-p)} \ne 0$ (since $p > q$ when $\varepsilon > 0$). By the implicit function theorem, $\frac{\partial p}{\partial \varepsilon} = -\frac{F_\varepsilon}{F_p} = \frac{1}{F_p} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = -\frac{F_q}{F_p} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. For the first-order certificate $B(\rho) = 2\,\overline{\mathrm{kl}}^{-1}(q(\rho), \varepsilon(\rho))$ with $q(\rho) = \rho^\top \widehat{L}$ and $\varepsilon(\rho) = \frac{\mathrm{KL}(\rho\|\pi)+c}{n}$, we have $\nabla_\rho q = \widehat{L}$ and $\nabla_\rho \varepsilon = \frac{1}{n}(\ln \frac{\rho}{\pi} + \mathbf{1})$, so by the chain rule
$$\nabla_\rho B = 2\Big[\frac{\partial p}{\partial q}\,\widehat{L} + \frac{\partial p}{\partial \varepsilon}\,\frac{1}{n}\Big(\ln \frac{\rho}{\pi} + \mathbf{1}\Big)\Big].$$
(With a softmax parametrization $\rho = \mathrm{softmax}(\theta)$, multiply by the Jacobian: $\nabla_\theta B = \rho \odot (g - \rho^\top g)$.) Numerically, at $q = 0.1$, $\varepsilon = 0.05$, $p \approx 0.220$: $\partial_\varepsilon p \approx 1.43$ and $\partial_q p \approx 1.33$.
5. The test-set bound certifies the vote itself, with a single kl inversion, no KL term, and no factor 2 or 4 coming from the Gibbs-to-vote relation; on a random forest

this relation is the main source of looseness. In exchange, the test examples are not used for learning, and a new split is needed for model selection. The self-bounding approach uses all the data, provides a learning objective with a guarantee, and allows model selection at the price of $\ln K$; with data-dependent priors it can be competitive when enough data are available.

Part III – Applications and short proofs

Exercise 13 – A majority vote by hand *(application)*

Five voters $h_1, \dots, h_5$ are evaluated on eight labelled points. Their predictions are:

point i	1	2	3	4	5	6	7	8
label y_i	+	−	+	+	−	+	−	−
$h_1(x_i)$	−	+	+	+	−	+	−	−
$h_2(x_i)$	−	−	−	+	−	+	−	−
$h_3(x_i)$	−	−	+	−	−	+	−	−
$h_4(x_i)$	+	−	+	+	+	−	−	−
$h_5(x_i)$	+	−	+	+	−	+	+	+

We treat the eight points as the whole distribution $\mathcal{D}$ (uniform on the eight points), so that all quantities below are exact ("oracle") values. Ties of the vote count as errors.

1. Give the risk of each voter. For the uniform posterior $Q = (\frac{1}{5}, \dots, \frac{1}{5})$, compute $W_Q(x_i, y_i)$ and the margin $M_Q = 1 - 2W_Q$ on each point, then $R_{\mathcal{D}}(G_Q)$ and $R_{\mathcal{D}}(B_Q)$.
2. Compute the joint error $e_Q = \mathbb{E}W_Q^2$ and the disagreement $d_Q = \mathbb{E}[2W_Q(1-W_Q)]$, and check that $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Compute the 5×5 matrix D of pairwise disagreements $D_{jk} = \mathbb{P}_x(h_j(x) \neq h_k(x))$ and check that $d_Q = \sum_{j,k} Q_j Q_k D_{jk}$. Which of these computations need the labels?
3. Compute the first-order bound $2R_{\mathcal{D}}(G_Q)$, the tandem bound $4e_Q$ and the C-bound $1 - \frac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_Q}$, and compare them with $R_{\mathcal{D}}(B_Q)$. Give the optimal parameter $\mu^\star$ of the Chebyshev–Cantelli family (Exercise 8, question 4) and check that it gives the C-bound.
4. Same questions for the posterior $Q' = (0.1, 0.1, 0.1, 0.35, 0.35)$. Why do the second-order bounds improve while the first-order bound does not move?
5. Same questions for the Dirac posterior $Q'' = \delta_{h_1}$. Which bound is the best now? Comment.

Solution

1. Comparing each row with the labels, h_1 errs on points $\{1, 2\}$, h_2 on $\{1, 3\}$, h_3 on $\{1, 4\}$, h_4 on $\{5, 6\}$ and h_5 on $\{7, 8\}$: every voter has risk $\frac{2}{8} = 0.25$. Under the uniform posterior, $W_Q(x_i, y_i)$ is the fraction of wrong voters: $W_Q = \frac{3}{5} = 0.6$ on point 1 (three errors) and $\frac{1}{5} = 0.2$ on the seven other points. Hence $M_Q = -0.2$ on point 1 and 0.6 elsewhere. The vote errs only on point 1: $R_{\mathcal{D}}(B_Q) = \frac{1}{8} = 0.125$, and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q = \frac{0.6+7\times0.2}{8} = 0.25$, the average of the individual risks as it should.
2. $e_Q = \frac{0.6^2+7\times0.2^2}{8} = \frac{0.36+0.28}{8} = 0.08$ and $d_Q = \frac{2\times0.6\times0.4+7\times2\times0.2\times0.8}{8} = \frac{0.48+2.24}{8} = 0.34$; indeed $0.08 + \frac{0.34}{2} = 0.25$. Two voters disagree exactly on the points where one of them errs and not the other. Among h_1, h_2, h_3, each pair shares the error

on point 1 and disagrees on two points ($D_{12} = D_{13} = D_{23} = 0.25$); h_4 and h_5 have errors disjoint from all the others, so any pair involving them disagrees on four points ($D_{jk} = 0.5$). With $Q_j Q_k = \frac{1}{25}$ and 20 ordered pairs $j \neq k$ (six inside $\{h_1, h_2, h_3\}$, fourteen others): $\sum_{j,k} Q_j Q_k D_{jk} = \frac{6\times 0.25 + 14 \times 0.5}{25} = \frac{8.5}{25} = 0.34 = d_Q$. The matrix D, hence d_Q, only compares rows of predictions: *no label is needed*. W_Q, e_Q, the risks and the vote risk need the labels.

3. First-order: $2 \times 0.25 = 0.5$. Tandem: $4 \times 0.08 = 0.32$. C-bound: $1 - \frac{(1-0.5)^2}{1-0.68} = 1 - \frac{0.25}{0.32} = 0.219$. All three are valid ($\geq 0.125$), and the C-bound is the tightest; the tandem bound improves on the first-order one because $d_Q = 0.34 \geq R_{\mathcal{D}}(G_Q) = 0.25$. The optimal parameter is $\mu^\star = \frac{R_{\mathcal{D}}(G_Q)/2 - e_Q}{1/2 - R_{\mathcal{D}}(G_Q)} = \frac{0.125 - 0.08}{0.25} = 0.18$, and $\frac{e_Q - 2\mu^\star R_{\mathcal{D}}(G_Q) + \mu^{\star 2}}{(1/2 - \mu^\star)^2} = \frac{0.08 - 0.09 + 0.0324}{0.1024} = \frac{0.0224}{0.1024} = 0.219$: the C-bound.

4. With Q': $W_{Q'} = 0.3$ on point 1, 0.1 on points 2–4, 0.35 on points 5–8, so all margins are positive (0.4, 0.8, 0.3) and $R_{\mathcal{D}}(B_{Q'}) = 0$. The Gibbs risk is unchanged, $R_{\mathcal{D}}(G_{Q'}) = \frac{0.3 + 3\times 0.1 + 4 \times 0.35}{8} = 0.25$ (all voters have the same risk), so the first-order bound stays at 0.5. But $e_{Q'} = \frac{0.09 + 3\times 0.01 + 4 \times 0.1225}{8} = \frac{0.61}{8} = 0.07625$ and $d_{Q'} = 2(0.25 - 0.07625) = 0.3475$ (check with D: $6 \times 0.01 \times 0.25 + 12 \times 0.035 \times 0.5 + 2 \times 0.1225 \times 0.5 = 0.015 + 0.21 + 0.1225 = 0.3475$). Tandem: 0.305; C-bound: $1 - \frac{0.25}{1 - 0.695} = 0.180$; $\mu^\star = \frac{0.125 - 0.07625}{0.25} = 0.195$. *Why:* h_1, h_2, h_3 all err on point 1; moving weight towards h_4, h_5, whose errors are not shared, decreases the joint error and increases the disagreement. The first-order bound only sees the average risk, which is the same for all voters.

5. With $Q'' = \delta_{h_1}$, $W \in \{0, 1\}$ (the vote is h_1): $R_{\mathcal{D}}(B_{Q''}) = R_{\mathcal{D}}(G_{Q''}) = 0.25$, $e = 0.25$, $d = 0$. First-order: 0.5; tandem: 1; C-bound: $1 - \frac{0.25}{1} = 0.75$. The first-order bound is now the best, as predicted by Exercise 1(j): without diversity, second-order bounds are useless.

What we learn: the benefit of the vote is invisible to first-order quantities; the joint error and the disagreement measure it, and the latter can be computed without labels. Even as oracle quantities the bounds remain pessimistic on such a tiny example (they are worst cases over all distributions of W_Q with given moments).

Exercise 14 – The Gibbs posterior on four voters *(application)*

Four voters have, on a sample of size $m = 200$, the empirical risks $\widehat{R}_S(h_1) = 0.10$, $\widehat{R}_S(h_2) = 0.12$, $\widehat{R}_S(h_3) = 0.15$, $\widehat{R}_S(h_4) = 0.30$ (i.e. 20, 24, 30 and 60 errors). The prior P is uniform and $\delta = 0.05$. Recall the Gibbs posterior $Q_\lambda(h) \propto P(h) e^{-\lambda m \widehat{R}_S(h)}$ (Exercise 5), Seeger's bound $R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}(\widehat{R}_S(G_Q), \frac{\mathrm{KL}(Q\|P) + \ln(2\sqrt{m}/\delta)}{m})$ and McAllester's bound $R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \sqrt{\frac{\mathrm{KL}(Q\|P) + \ln(2\sqrt{m}/\delta)}{2m}}$. You may use `kl_inv_upper` of `pbtools` for the inversions.

1. Show that $Q_\lambda(h_j) = \frac{e^{-\lambda(k_j - k_1)}}{\sum_l e^{-\lambda(k_l - k_1)}}$, where k_j is the number of errors of h_j. Why is this form numerically preferable? Compute Q_λ for $\lambda \in \{0.05, 0.25, 1\}$.

2. For $\lambda \in \{0, 0.05, 0.25, 1\}$ and for $\lambda \to +\infty$, compute $\widehat{R}_S(G_{Q_\lambda})$, $\mathrm{KL}(Q_\lambda\|P)$ (use $\mathrm{KL} = \ln 4 - H(Q_\lambda)$, Exercise 4), and both bounds.

3. For $\lambda = 0.25$, compute the objective $\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda m}$ for Q_λ, for the uniform posterior and for δ_{h_1}, and check the value $-\frac{1}{\lambda m} \ln \mathbb{E}_{h\sim P} e^{-\lambda m \widehat{R}_S(h)}$ found in Exercise 5.

4. Which posterior gives the best certificate? Is it legitimate to choose λ after computing the bounds? Why does the bound favour concentrated posteriors here, and what would this

imply for the majority vote?

Solution

1. $Q_\lambda(h_j) = \frac{\frac{1}{4}e^{-\lambda k_j}}{\sum_l \frac{1}{4}e^{-\lambda k_l}}$ since $m\widehat{R}_S(h_j) = k_j$; multiplying numerator and denominator by $e^{\lambda k_1}$ gives the formula. The largest exponential is now $e^0 = 1$: no underflow even when λk_j is large (with $m = 10^5$, $e^{-\lambda k_j}$ would be 0 in floating point). This is the "log-sum-exp" trick used by `softmax` in `pbtools`. With $k_j - k_1 = (0, 4, 10, 40)$:

λ	unnormalized weights	sum	Q_λ
0.05	$(1,\ 0.819,\ 0.607,\ 0.135)$	2.561	$(0.391,\ 0.320,\ 0.237,\ 0.053)$
0.25	$(1,\ 0.368,\ 0.082,\ 4.5\cdot10^{-5})$	1.450	$(0.690,\ 0.254,\ 0.057,\ 0.000)$
1	$(1,\ 0.018,\ 4.5\cdot10^{-5},\ 4\cdot10^{-18})$	1.018	$(0.982,\ 0.018,\ 0.000,\ 0.000)$

2. Here $\ln\frac{2\sqrt{200}}{0.05} = \ln 565.7 = 6.338$ and $\varepsilon = \frac{\mathrm{KL}+6.338}{200}$. For $\lambda = 0.05$: $\widehat{R}_S(G_{Q_\lambda}) = 0.391\times0.10 + 0.320\times0.12 + 0.237\times0.15 + 0.053\times0.30 = 0.1288$, $H(Q_\lambda) = 1.228$, $\mathrm{KL} = 1.386 - 1.228 = 0.158$. The other columns are obtained in the same way:

	$\lambda = 0$ (uniform)	0.05	0.25	1	∞ (δ_{h_1})
$\widehat{R}_S(G_{Q_\lambda})$	0.1675	0.1288	0.1079	0.1004	0.1000
$\mathrm{KL}(Q_\lambda\|P)$	0	0.158	0.619	1.296	1.386
ε	0.0317	0.0325	0.0348	0.0382	0.0386
Seeger	0.274	0.229	0.207	0.203	0.203
McAllester	0.293	0.256	0.240	0.239	0.239

   (The Dirac on h_1 has $\mathrm{KL} = \ln 4$, the Occam price of selecting one voter out of four.) Seeger's bound is always tighter than McAllester's, by 0.02 to 0.036, the gap growing as the posterior concentrates on voters with small empirical risk (the risks are far from $\frac{1}{2}$).

3. With $\lambda m = 50$: for $Q_{0.25}$, $0.1079 + \frac{0.619}{50} = 0.1203$; for the uniform posterior, $0.1675 + 0 = 0.1675$; for δ_{h_1}, $0.1 + \frac{1.386}{50} = 0.1277$. The Gibbs posterior is indeed the smallest, and $-\frac{1}{50}\ln\left(\frac{1}{4}(e^{-5}+e^{-6}+e^{-7.5}+e^{-15})\right) = 0.1203$ coincides with its value, as predicted by the variational formula (it is a soft minimum of the empirical risks, between 0.10 and their mean).

4. The bound decreases from 0.274 (uniform) to about 0.203 ($\lambda \geq 1$); a finer scan gives a flat minimum 0.2028 around $\lambda \approx 0.8$, barely better than the Dirac. Choosing λ after computing the bounds is *free*: λ only indexes posteriors, and Seeger's bound holds for all posteriors simultaneously (Exercise 12, question 3(a)). With only four voters, the KL can never exceed $\ln 4 = 1.39$ nats, small compared with the constant 6.34, so concentrating on the best voter is cheap and it reduces the empirical term by 0.07. The resulting "vote" is essentially h_1 alone: all diversity is lost, which is the phenomenon of Exercise 10, question 4. Minimizing a Gibbs-risk bound is the right objective for the Gibbs classifier, not for the vote.

Exercise 15 – Data-dependent priors and sample splitting *(short proof and application)*

We learn a linear classifier with a Gaussian posterior $Q = \mathcal{N}(\mathbf{w}, I_d)$ from a sample S of size $m = 1000$, $\delta = 0.05$. To obtain a better prior, we split $S = S_1 \cup S_2$ with $|S_1| = m_1$ and $|S_2| = m_2 = m - m_1$, learn a vector $\mathbf{w}_{S_1}$ on S_1 only ($\mathbf{w}_{S_1} = \mathbf{0}$ if $m_1 = 0$), and use the prior $P_{S_1} = \mathcal{N}(\mathbf{w}_{S_1}, I_d)$. The posterior centre $\mathbf{w}$ is learned on the whole sample S.

1. *(Short proof.)* Show that, with probability at least $1-\delta$ over (S_1, S_2), for all posteriors Q, $R_{\mathcal{D}}(G_Q) \le \overline{\mathrm{kl}}^{-1}(\widehat{R}_{S_2}(G_Q), \frac{\mathrm{KL}(Q\|P_{S_1}) + \ln(2\sqrt{m_2}/\delta)}{m_2})$. Why may Q depend on the whole of S, while the empirical risk must be computed on S_2 only?
2. The following values were observed for four splits (in all cases $\widehat{R}_{S_2}(G_Q) = 0.10$):

m_1	0	200	500	800
$\|\mathbf{w} - \mathbf{w}_{S_1}\|^2$	120	30	10	4

   Compute the KL (Exercise 4) and the certificate for each split.
3. Discuss the trade-off governing the choice of m_1.
4. An analyst computes the four certificates and reports the smallest one. What should she report instead?
5. What does the bound become if the posterior is the prior itself, $Q = P_{S_1}$? Relate it to the test-set bound of Exercise 3.

Solution

1. Condition on S_1. Then P_{S_1} is a fixed distribution, and $S_2 \sim \mathcal{D}^{m_2}$ is independent of S_1, hence of P_{S_1}. Seeger's bound applied to the sample S_2 with the prior P_{S_1} gives, for every fixed value of S_1, $\mathbb{P}_{S_2}(E \,|\, S_1) \ge 1-\delta$, where E is the event "for all Q, the inequality holds". Integrating over S_1, $\mathbb{P}(E) = \mathbb{E}_{S_1}[\mathbb{P}_{S_2}(E \,|\, S_1)] \ge 1-\delta$. Since on E the inequality holds *for all* Q, it holds for a posterior learned on $S_1 \cup S_2$. The empirical term, on the contrary, is the variable that concentrates in the proof: its exponential moment is computed under $\mathbb{E}_{S_2}\mathbb{E}_{h\sim P_{S_1}}$, which requires S_2 to be independent of the prior. The empirical risk on S_1 is biased in favour of the voters that P_{S_1} puts forward (they were chosen because they fit S_1), so using S_1 in the empirical term would break the proof.
2. By Exercise 4.4 with $\sigma = 1$, $\mathrm{KL} = \frac{1}{2}\|\mathbf{w} - \mathbf{w}_{S_1}\|^2$. With $c = \ln(2\sqrt{m_2}/\delta)$:

m_1	m_2	KL	c	$\varepsilon = \frac{\mathrm{KL}+c}{m_2}$	$\overline{\mathrm{kl}}^{-1}(0.10, \varepsilon)$
0	1000	60	7.14	0.0671	0.243
200	800	15	7.03	0.0275	0.185
500	500	5	6.80	0.0236	**0.177**
800	200	2	6.34	0.0417	0.208

3. Increasing m_1 gives a better prior, closer to the posterior, hence a smaller KL; but it leaves fewer examples m_2 to estimate the empirical risk and to divide the complexity term. Without splitting, the KL (60 nats) dominates; with $m_1 = 800$ the KL is almost gone but ε is limited by $\frac{c}{m_2} \approx 0.03$. The best trade-off is intermediate ($m_1 = 500$ here), as in Figure 28 of the notes. Splitting helps when the KL of the data-free prior is large compared with $\ln(2\sqrt{m}/\delta)$, as for large models.
4. Choosing the split with the data is a choice among $K = 4$ bounds (different priors

and samples): by the union bound (Proposition 6.3 of the notes), each certificate must be computed with $\delta/4$, i.e. with $\ln(8\sqrt{m_2}/\delta)$. This gives 0.245, 0.188, 0.182 and 0.218: she can report 0.182 for $m_1 = 500$. The price, $\ln 4 = 1.4$ nats, is small, but it must be paid (and the list of candidate splits must be fixed before looking at the data).

5. With $Q = P_{S_1}$, KL $= 0$ and the certificate is $\overline{\mathrm{kl}}^{-1}(\widehat{R}_{S_2}(G_{P_{S_1}}), \frac{\ln(2\sqrt{m_2}/\delta)}{m_2})$: it is a test-set bound on S_2 for the (stochastic) classifier $G_{P_{S_1}}$, which was fixed before seeing S_2. It is slightly looser than the test-set bound of Exercise 3 ($\ln(2\sqrt{m_2}/\delta)$ instead of $\ln(1/\delta)$) because it also covers all the other posteriors. The two ends of the spectrum are thus pure hold-out ($Q = P_{S_1}$) and pure self-bounding ($m_1 = 0$).

Exercise 16 – Certifying the balanced error *(short proof and application)*

For a classifier f, let $R_+(f) = \mathbb{P}(f(x) \neq y \,|\, y = +1)$ and $R_-(f) = \mathbb{P}(f(x) \neq y \,|\, y = -1)$ be the class-conditional risks, $\mathrm{BER}(f) = \frac{1}{2}(R_+(f) + R_-(f))$ the balanced error, and $\pi_+ = \mathbb{P}(y = +1)$, so that $R_{\mathcal{D}}(f) = \pi_+ R_+(f) + (1 - \pi_+)R_-(f)$. A classifier f, learned on another sample, is evaluated on an independent test sample of $n = 1000$ examples with $n_+ = 100$ positives and $n_- = 900$ negatives; it makes 12 errors on the positives and 18 on the negatives.

1. Compute the kl test-set bound on $R_{\mathcal{D}}(f)$. Why is this certificate misleading for this problem?

2. *(Short proof.)* Conditionally on the labels $(y_1, \dots, y_n)$, show that $n_+\widehat{R}_+(f) \sim \mathrm{Bin}(n_+, R_+(f))$, where $\widehat{R}_+(f)$ is the error rate on the positive test examples. Deduce that with probability at least $1 - \delta$, simultaneously, $R_+(f) \leq U_+ := \overline{\mathrm{kl}}^{-1}(\widehat{R}_+(f), \frac{\ln(2/\delta)}{n_+})$ and $R_-(f) \leq U_- := \overline{\mathrm{kl}}^{-1}(\widehat{R}_-(f), \frac{\ln(2/\delta)}{n_-})$; hence $\mathrm{BER}(f) \leq \frac{U_+ + U_-}{2}$ and $R_{\mathcal{D}}(f) \leq \max(U_+, U_-)$.

3. Compute U_+, U_- and the certificate on the balanced error. What does it add to question 1? What could you say about $R_{\mathcal{D}}(f)$ if $\pi_+ = 0.1$ were known?

4. ($\star$) *(Another route.)* Conditionally on the labels, write the empirical balanced error as a sum $\sum_i Z_i$ of independent variables with $Z_i \in [0, c_i]$, $c_i = \frac{1}{2n_{y_i}}$. Adapting the Chernoff method of Exercise 2, show that $\mathrm{BER}(f) \leq \widehat{\mathrm{BER}}(f) + \sqrt{\frac{\ln(1/\delta)}{8}(\frac{1}{n_+} + \frac{1}{n_-})}$ with probability $\geq 1 - \delta$. Compute it and compare with question 3.

5. How would you obtain a PAC-Bayesian certificate on the balanced error of the Gibbs classifier G_Q, valid for all posteriors Q? And of the majority vote?

Solution

1. $\widehat{R}_T(f) = \frac{30}{1000} = 0.03$ and $\overline{\mathrm{kl}}^{-1}(0.03, \frac{\ln 20}{1000}) = \overline{\mathrm{kl}}^{-1}(0.03, 0.0030) = 0.045$. The certificate is correct but dominated by the 900 negatives: it says nothing about the minority class, on which f errs 12% of the time. A classifier predicting -1 everywhere would have a test error of 0.10 and a certificate of $\overline{\mathrm{kl}}^{-1}(0.1, 0.0030) = 0.125$, while being useless for the positives.

2. Given the labels, the pairs are independent and $x_i \,|\, y_i \sim \mathcal{D}_{\mathcal{X}|Y=y_i}$. For the n_+ indices with $y_i = +1$, the indicators $\mathbb{1}[f(x_i) \neq +1]$ are thus i.i.d. Bernoulli with parameter $\mathbb{P}(f(x) \neq y \,|\, y = +1) = R_+(f)$ (f is fixed, learned on an independent sample); their sum is $\mathrm{Bin}(n_+, R_+(f))$. The test-set bound (Chernoff) with confidence $\delta/2$ gives $\mathbb{P}(R_+(f) > U_+ \,|\, y_{1:n}) \leq \frac{\delta}{2}$, and similarly for the negatives. By the union bound, both inequalities hold with conditional probability $\geq 1 - \delta$, for every value of the labels, hence also unconditionally. On this event, $\mathrm{BER} = \frac{R_+ + R_-}{2} \leq \frac{U_+ + U_-}{2}$, and

$R_{\mathcal{D}}(f) = \pi_+ R_+ + (1-\pi_+)R_-$ is a convex combination of R_+ and R_-, hence at most $\max(U_+, U_-)$.

3. $\widehat{R}_+ = 0.12$, $\varepsilon_+ = \frac{\ln 40}{100} = 0.0369$, $U_+ = \overline{\mathrm{kl}}^{-1}(0.12, 0.0369) = 0.226$; $\widehat{R}_- = 0.02$, $\varepsilon_- = \frac{\ln 40}{900} = 0.0041$, $U_- = \overline{\mathrm{kl}}^{-1}(0.02, 0.0041) = 0.035$. Hence $\mathrm{BER}(f) \le \frac{0.226+0.035}{2} = 0.131$ (empirical value 0.07). The certificate reveals that the error on the positives may be as large as 23%, which the global certificate 0.045 hid; the uncertainty is dominated by the small class ($n_+ = 100$). If $\pi_+ = 0.1$ is known, $R_{\mathcal{D}}(f) \le 0.1 \times 0.226 + 0.9 \times 0.035 = 0.054$, slightly worse than 0.045 (we paid $\delta/2$ and the small class twice), but together with a guarantee per class.

4. $\widehat{\mathrm{BER}} = \frac{1}{2n_+}\sum_{y_i=+1}\ell_i + \frac{1}{2n_-}\sum_{y_i=-1}\ell_i = \sum_i Z_i$ with $Z_i = \frac{\ell_i}{2n_{y_i}} \in [0, c_i]$, independent given the labels, and $\mathbb{E}\sum_i Z_i = \mathrm{BER}$. For $\lambda > 0$, by Markov and Hoeffding's lemma applied to each $\mathbb{E}Z_i - Z_i$ (range c_i),

$$\mathbb{P}(\mathrm{BER} - \widehat{\mathrm{BER}} \ge t) \le e^{-\lambda t}\prod_i \mathbb{E}e^{\lambda(\mathbb{E}Z_i - Z_i)} \le \exp\Big(-\lambda t + \frac{\lambda^2}{8}\sum_i c_i^2\Big).$$

   The choice $\lambda = 4t/\sum_i c_i^2$ gives $\exp(-2t^2/\sum_i c_i^2)$, and $\sum_i c_i^2 = \frac{n_+}{4n_+^2} + \frac{n_-}{4n_-^2} = \frac{1}{4}(\frac{1}{n_+} + \frac{1}{n_-})$. Setting the probability to δ gives $t = \sqrt{\frac{\ln(1/\delta)}{8}(\frac{1}{n_+} + \frac{1}{n_-})}$. Numerically, $t = \sqrt{\frac{2.996}{8} \times 0.0111} = 0.0645$ and the bound is $0.07 + 0.0645 = 0.135$, slightly worse than 0.131: it avoids the union bound but uses Hoeffding instead of kl, which is costly for the small risk of the negative class.

5. Apply Seeger's bound separately to the positive and negative sub-samples, each with confidence $\delta/2$ (and the same prior P): with probability $\ge 1-\delta$, for all Q, $R_\pm(G_Q) \le \overline{\mathrm{kl}}^{-1}(\widehat{R}_\pm(G_Q), \frac{\mathrm{KL}(Q\|P) + \ln(4\sqrt{n_\pm}/\delta)}{n_\pm})$, and the balanced Gibbs risk is at most the average. For the vote, the first-order argument works class by class: $R_+(B_Q) = \mathbb{P}(W_Q \ge \frac{1}{2} \,|\, y = +1) \le 2\mathbb{E}[W_Q \,|\, y = +1] = 2R_+(G_Q)$, and similarly for R_-, so $\mathrm{BER}(B_Q) \le 2\,\mathrm{BER}(G_Q)$.

What we learn: a certificate is about one specific risk; for imbalanced problems, certifying each class separately (union bound) costs little and is much more informative.

Exercise 17 – Softmax parametrization and the gradient of a certificate *(short proofs and application)*

A posterior ρ on n_v voters is parametrized by logits $\theta \in \mathbb{R}^{n_v}$: $\rho = \mathrm{softmax}(\theta)$, i.e. $\rho_t = e^{\theta_t}/\sum_s e^{\theta_s}$. Let π be the prior, $\widehat{L} \in [0,1]^{n_v}$ the vector of empirical losses, $\mathcal{B}(\rho)$ a differentiable objective and $g = \nabla_\rho \mathcal{B}$.

1. Show that $\frac{\partial \rho_t}{\partial \theta_s} = \rho_t(\mathbb{1}[t=s] - \rho_s)$, i.e. the Jacobian is $\mathrm{diag}(\rho) - \rho\rho^\top$.

2. Deduce $\nabla_\theta \mathcal{B} = \rho \odot (g - (\rho^\top g)\mathbf{1})$. Show that $\mathbf{1}^\top \nabla_\theta \mathcal{B} = 0$ and explain this with the invariance $\mathrm{softmax}(\theta + c\mathbf{1}) = \mathrm{softmax}(\theta)$. Why can g be modified by adding a multiple of $\mathbf{1}$?

3. Show that $\nabla_\rho \mathrm{KL}(\rho\|\pi) = \ln\frac{\rho}{\pi} + \mathbf{1}$ and $\nabla_\theta \mathrm{KL}(\rho\|\pi) = \rho \odot (\ln\frac{\rho}{\pi} - \mathrm{KL}(\rho\|\pi)\mathbf{1})$. What is this gradient at $\rho = \pi$?

4. Let $F(\rho) = \rho^\top \widehat{L} + \frac{\mathrm{KL}(\rho\|\pi)}{\lambda n}$. Show that $\nabla_\theta F = 0$ iff ρ is the Gibbs posterior $\rho_t \propto \pi_t e^{-\lambda n \widehat{L}_t}$.

5. Consider the first-order certificate $\mathcal{B}(\rho) = 2\,\overline{\mathrm{kl}}^{-1}(q(\rho), \varepsilon(\rho))$ with $q(\rho) = \rho^\top \widehat{L}$ and $\varepsilon(\rho) = \frac{\mathrm{KL}(\rho\|\pi) + \ln(2\sqrt{n}/\delta)}{n}$. Admitting the derivatives $\frac{\partial p}{\partial q}$ and $\frac{\partial p}{\partial \varepsilon}$ of $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ (Exercise 12,

question 4), write $\nabla_\theta \mathcal{B}$, and simplify it at $\rho = \pi$.

6. Numerical application: three voters, $\widehat{L} = (0.1, 0.2, 0.3)$, π uniform, $\theta = \mathbf{0}$, $n = 200$, $\delta = 0.05$. Compute ε, $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ (with `kl_inv_upper`), $\mathcal{B}$, the two partial derivatives and $\nabla_\theta \mathcal{B}$. Perform one gradient step $\theta \leftarrow \theta - 5\nabla_\theta \mathcal{B}$ and compute the new certificate. Where do repeated steps lead?

Solution

1. Let $Z = \sum_s e^{\theta_s}$, so $\partial Z / \partial \theta_s = e^{\theta_s}$. By the quotient rule, $\frac{\partial \rho_t}{\partial \theta_s} = \frac{\mathbb{1}[t=s] e^{\theta_t} Z - e^{\theta_t} e^{\theta_s}}{Z^2} = \mathbb{1}[t = s]\, \rho_t - \rho_t \rho_s$. In matrix form $J = \mathrm{diag}(\rho) - \rho\rho^\top$, which is symmetric.

2. By the chain rule, $\frac{\partial \mathcal{B}}{\partial \theta_s} = \sum_t g_t \frac{\partial \rho_t}{\partial \theta_s} = \rho_s g_s - \rho_s \sum_t \rho_t g_t = \rho_s (g_s - \rho^\top g)$, which is $\nabla_\theta \mathcal{B} = J^\top g = \rho \odot (g - (\rho^\top g)\mathbf{1})$. Summing over s: $\mathbf{1}^\top \nabla_\theta \mathcal{B} = \rho^\top g - (\rho^\top g) \sum_s \rho_s = 0$. Indeed $\theta \mapsto \mathcal{B}(\mathrm{softmax}(\theta))$ is constant along the direction $\mathbf{1}$, so its derivative in that direction vanishes. Replacing g by $g + c\mathbf{1}$ adds $c\rho \odot (\mathbf{1} - \mathbf{1}) = 0$: only the variations of $\mathcal{B}$ *within* the simplex matter, and g is defined up to the direction $\mathbf{1}$ (which is why constant terms such as the $+\mathbf{1}$ of the KL gradient are harmless).

3. $\mathrm{KL}(\rho\|\pi) = \sum_t \rho_t \ln \rho_t - \rho_t \ln \pi_t$, so $\frac{\partial\, \mathrm{KL}}{\partial \rho_t} = \ln \rho_t + 1 - \ln \pi_t$. Then $\rho^\top g = \mathrm{KL}(\rho\|\pi) + 1$, and question 2 gives $\nabla_\theta \,\mathrm{KL} = \rho \odot \big(\ln \frac{\rho}{\pi} + \mathbf{1} - (\mathrm{KL} + 1)\mathbf{1}\big) = \rho \odot \big(\ln \frac{\rho}{\pi} - \mathrm{KL}\ \mathbf{1}\big)$. At $\rho = \pi$, $\ln \frac{\rho}{\pi} = \mathbf{0}$ and $\mathrm{KL} = 0$: the gradient vanishes. The KL is minimal at the prior, so moving slightly away from π costs nothing at first order.

4. Here $g = \widehat{L} + \frac{1}{\lambda n}\big(\ln \frac{\rho}{\pi} + \mathbf{1}\big)$, so by questions 2–3, $\nabla_\theta F = \rho \odot \big(\widehat{L} - (\rho^\top \widehat{L})\mathbf{1} + \frac{1}{\lambda n}(\ln \frac{\rho}{\pi} - \mathrm{KL}\ \mathbf{1})\big)$. Since all $\rho_t > 0$, this vanishes iff $\widehat{L}_t + \frac{1}{\lambda n} \ln \frac{\rho_t}{\pi_t}$ does not depend on t, i.e. $\ln \frac{\rho_t}{\pi_t} = -\lambda n \widehat{L}_t + \text{const}$, i.e. $\rho_t \propto \pi_t e^{-\lambda n \widehat{L}_t}$. We recover the Gibbs posterior of Exercise 5 without Lagrange multiplier: the softmax parametrization handles the constraint automatically.

5. With $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$, the chain rule gives $g = \nabla_\rho \mathcal{B} = 2\big[\frac{\partial p}{\partial q}\widehat{L} + \frac{\partial p}{\partial \varepsilon}\frac{1}{n}(\ln \frac{\rho}{\pi} + \mathbf{1})\big]$, and by questions 2–3,

$$\nabla_\theta \mathcal{B} = 2\Big[\frac{\partial p}{\partial q}\, \rho \odot (\widehat{L} - q\mathbf{1}) + \frac{\partial p}{\partial \varepsilon}\, \frac{1}{n}\, \rho \odot \Big(\ln \frac{\rho}{\pi} - \mathrm{KL}(\rho\|\pi)\mathbf{1}\Big)\Big].$$

   At $\rho = \pi$ the second term vanishes and $\nabla_\theta \mathcal{B} = 2\frac{\partial p}{\partial q}\, \pi \odot (\widehat{L} - q\mathbf{1})$: the logits of the voters better than average increase, the others decrease.

6. $q = 0.2$, $\mathrm{KL} = 0$, $\ln \frac{2\sqrt{200}}{0.05} = 6.338$, $\varepsilon = 0.0317$, $p = \overline{\mathrm{kl}}^{-1}(0.2, 0.0317) = 0.312$ and $\mathcal{B} = 0.624$. Then $\frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q} = \frac{0.312 \times 0.688}{0.112} = 1.92$ and $\frac{\partial p}{\partial q} = 1.92 \times \ln \frac{p(1-q)}{q(1-p)} = 1.92 \times \ln 1.813 = 1.92 \times 0.595 = 1.14$. Hence $\nabla_\theta \mathcal{B} = 2 \times 1.14 \times \frac{1}{3}(-0.1, 0, 0.1) = (-0.076, 0, 0.076)$ (this agrees with `certificate_and_grad` of `pbtools`). After the step, $\theta = (0.380, 0, -0.380)$, $\rho = (0.465, 0.318, 0.217)$, $q = 0.175$, $\mathrm{KL} = \ln 3 - H(\rho) = 0.047$, $\varepsilon = 0.0319$ and $\mathcal{B} = 2\overline{\mathrm{kl}}^{-1}(0.175, 0.0319) = 0.568$: the certificate decreased by 0.056, close to the first-order prediction $5\|\nabla_\theta \mathcal{B}\|^2 = 0.058$. Repeated steps keep moving the weight towards h_1, until the growth of the KL balances the decrease of q: the first-order certificate favours concentrated posteriors (Exercises 10 and 14). For the tandem certificate, the gradient contains $2\widehat{\mathbf{L}}\rho$ instead of $\widehat{L}$, which penalizes voters that err together and preserves diversity.